

THEA – THE LIGHT MATRIX

Calculation-by-Calculation LaTeX Tower v1.3.3

exact closure, golden selection, chirality, spectra, invariant theory, numerical walls, and the
LUCA handoff

Independent audit edition

Buenos Aires + Ancient Korinthos lineage

Prepared 9 August 2026 - expanded calculation audit

Abstract

This is the calculation-by-calculation successor to the frozen v1.3.2 audit. It retains the full narrative tower and adds one explicit rung for every independently checked calculation: 38 topology, lattice, light-matrix, spectral, invariant-theory, precision, and LUCA boundary checks; a source replay of every LUCA v0.2 primitive; and a complete mathematical autopsy of the newly supplied *THE CROWD* coincidence engine.

The exact core remains unchanged: the frozen verifier returns 38/38 passes; the unmodified Goldberg/Lanczos source reproduces eight closed-shell spectral receipts; and the unmodified LUCA v0.2 source locks 34/34 non-axiomatic calculations, with 44,185 EML calls and a worst row still above twelve correct digits.

The new statistical audit reproduces every table printed by *THE CROWD*. Its apparent 3.0σ row is a marginal result selected from five nested tiers. The empirical maximum-over-tier correction gives $p = 0.0215$ (about 2.02 one-sided Gaussian standard deviations). Removing the shared mathematical alphabet gives $p = 0.0950$; the source's seven-representative deduplication gives $p = 0.3270$; a cluster-preserving matched null gives $p = 0.1090$; and a strict four-generator provenance collapse gives $p = 0.2590$. No physical identification survives.

A limited 2026 PDG re-pin updates the W, Higgs, and direct top masses that drive the strongest electroweak nearest neighbours. It changes individual hit rankings but not the post-autopsy verdict. This document therefore preserves the governing rule in executable form: target is not result; a theorem, a finite computation, a design choice, a null result, and a physical hypothesis never inherit one another's labels.

**The pentagons hold. The hexes pay. The matrix
remembers which is which.**

$$P = 12, \quad \chi = 2, \quad \text{spec}(\mathcal{M}_{\text{light}}) = \{\varphi^2, 1, -1, \varphi^{-2}\}.$$

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Table Map

The principal tabular receipts are intentionally embedded in the mathematical narrative rather than separated from their hypotheses. Their locations are:

Source inventory

Chapter 1: the seven audited artifacts and their roles.

Golden shell catalogue

Chapter 4: (k_n, ℓ_n) , T_n , cage size, hexagon count, and chirality angle.

Light-matrix eigenbasis

Chapter 5: all four modes and their exact eigenvectors.

C_{60} **polynomial** Chapter 6: the complete adjacency characteristic factorization.

Live spectral receipt

Chapter 7: the v1.3.1 Goldberg-builder/Lanczos output through C_{17660} .

Precision formats

Chapter 10: significand budgets and exact-integer scale limits.

LUCA comparisons

Chapter 11: finite- N angle sweep and the 32/33-point leaf-regression distinction.

Independent audit

Chapter 14: all 38 checks, statuses, verdicts, and compact results; full values remain in the JSON receipt.

Chapter 1

The Contract: Status Before Symbol

1.1 Source inventory and lineage

Artifact	file	Use in this tower
THEA core	Thea.md	Exact topology, closure algebra, Fibonacci selector, light matrix, spectral receipts, and the honest physical boundary.
Light Matrix shell	shell__thea_light_matrix_v1.3.1.html	Nineteen interactive sections, exact C_{60} kernel, Goldberg builder, spectral routines, $A_2/SU(3)$ lane, Klein lane, and numerical-architecture lane.
LUCA v0.1	shell__eml_luca_spiral_v0_1.html	EML operator, reduction tree, branch receipts, and target/current/error verification.
LUCA v0.2	shell__eml_luca_spiral_v0_2.html	Live Faddeev–LeVerrier factorization, golden selector readout, golden-ray angle, finite- N Vogel sweep, and leaf-count null result.
Angle handoff	THE_ANGLE_OF_THE_GOLDEN_RAY.md	Radical angle, asymptotic alternating convergence, anti-palindromic characteristic polynomial, and build contract.
Twelve Paths	THE_12_PATHS_OF_THE_FRACTAL_MAGE.md	Proof by kernel, target $\neq$ result, immutable versions, and open receipts.
Kernel grimoire	KERNELIMAGIC.md	False convergence, pre-allocation cost gates, deterministic certificates, and portable paths.
THE CROWD scroll	THE_CROWD.md	Coincidence scan, declared null, shared-alphabet autopsy, deduplication, precision ladder, and sealed prediction protocol.
THE CROWD kernel	the_crowd_v0_1.py	Deterministic seed-20260809 engine whose tables and Monte Carlo null are replayed independently.

1.2 The status grammar

Every displayed claim in v1.3.3 belongs to one of the following classes.

EXACT Follows from algebra, topology, integer arithmetic, or a symbolic identity shown in the document.

COMPUTED Reproduced by a named finite algorithm at a stated precision, sample count, graph depth, or iteration count.

DESIGN CHOICE A visualization, parameterization, tolerance, mapping, sampling schedule, or user-interface rule.

HYPOTHESIS A proposed physical interpretation that still requires discriminating evidence.

EXTERNAL / METAPHOR A statement inherited from another artifact or source and not independently proved here.

CORRECTION A source statement whose algebra, label, indexing, or implementation receipt changes in v1.3.3.

v1.3.3 correction

The HTML header advertises additions through Section XVII, yet the baked document contains a Section XVIII; the HUD count of 19 includes the birth section plus XVIII. v1.3.2 makes the section inventory explicit. It also repairs the abstract glyph C_{00} to C_{60} and replaces “zero dependencies” by the narrower, accurate statement that the mathematical kernel is dependency-free while presentation may use KaTeX.

1.3 Audit outcome in one page

The independent Python/SymPy/NetworkX/NumPy audit executed 38 checks and returned 38 passes. “Pass” here means that either the source statement was reproduced or its correction was independently established. The most important changes are:

1. The sequence T_{n+1}/T_n does not monotonically climb to φ^2 ; it alternates around the limit.
2. Golden-selected shells form a catalogue of independently closed shells. They are not an exactly nested fixed Goldberg transform with linear multiplier φ .
3. Section VIII applies Lanczos to a graph Laplacian, not to the 4×4 light matrix.
4. The continuum coefficient $2\pi/(5\sqrt{3})$ is exact conditional algebra. The graph-limit arrow remains a conjectural asymptotic identification supported by finite computation.
5. $Q(A_2)$ and $P(A_2)$ are not literally equal. The common hexagonal quadratic form must be stated with normalization.
6. The Klein syzygy is exact; basin counts are finite, seed-dependent numerical coverage experiments.
7. Binary256 has 237 bits of significand precision, not a “256-bit mantissa.”
8. The v1.3.1 recurrence comparator shared the same binary64 rounding in both operands. A rational comparison against the exact integer gives a true maximum relative error of about 1.02×10^{-15} over levels 0–159.
9. LUCA branch identities require domains and branch sheets. A sample lock is a sieve, not a proof.
10. The live LUCA v0.2 regression uses 33 spiral nodes; the 32-point handoff values are recovered only after excluding e .

v1.3.3 statistical correction

The newly supplied coincidence engine reproduces exactly, but its 3.0σ row is marginal. Taking the maximum over the five scanned tiers gives empirical global $p = 0.0215$. The source's own physics-only and deduplication autopsies then give $p = 0.0950$ and $p = 0.3270$. A new cluster-preserving matched null gives $p = 0.1090$, and a stricter four-generator provenance collapse gives $p = 0.2590$. The crowd identifies nobody.

Chapter 2

Closed Topology: Why Twelve Cannot Move

2.1 General trivalent curvature ledger

Let f_p be the number of p -gonal faces in a connected, closed, trivalent planar graph embedded on S^2 . Then

$$3V = 2E, \quad \sum_{p \geq 3} p f_p = 2E, \quad V - E + \sum_{p \geq 3} f_p = 2. \quad (2.1)$$

Eliminating V and E gives the exact combinatorial curvature identity

$$\boxed{\sum_{p \geq 3} (6 - p) f_p = 12.} \quad (2.2)$$

This is more general than the fullerene statement. Triangles contribute +3, squares +2, pentagons +1, hexagons zero, and larger faces negative charge.

2.2 The fullerene specialization

For a pentagon/hexagon fullerene, $f_5 = P$, $f_6 = H$, and every other f_p vanishes. Equation (2.2) becomes

$$(6 - 5)P + (6 - 6)H = 12, \quad (2.3)$$

so

Exact statement

$$\boxed{P = 12.} \quad (2.4)$$

The assumptions are part of the theorem: connected, closed, spherical, trivalent, and containing only pentagonal and hexagonal faces.

The remaining counts follow:

$$\boxed{E = \frac{3V}{2}, \quad H = \frac{V}{2} - 10, \quad F = P + H = \frac{V}{2} + 2.} \quad (2.5)$$

For C_{60} ,

$$(V, E, F, P, H, \chi) = (60, 90, 32, 12, 20, 2). \quad (2.6)$$

2.3 Discrete Gauss–Bonnet

Assign the face charge

$$K_p = \frac{\pi}{3}(6 - p). \quad (2.7)$$

Then

$$\sum_f K_f = \frac{\pi}{3} \sum_p (6 - p) f_p = \frac{\pi}{3} \cdot 12 = 4\pi. \quad (2.8)$$

Honest boundary

The equality is exact combinatorial topology. It does not assert that a molecular fullerene carries uniform smooth Gaussian curvature, nor that graph curvature is physical spacetime curvature. It is a ledger of angular defect for this trivalent tiling class.

Chapter 3

The Hexagonal Closure Algebra

3.1 The sixth-root coordinate

Set

$$\zeta_6 = e^{i\pi/3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}, \quad \zeta_6^2 = \zeta_6 - 1. \quad (3.1)$$

A lattice displacement is

$$z = k + \ell\zeta_6, \quad k, \ell \in \mathbb{Z}. \quad (3.2)$$

Its squared Euclidean length is

$$\boxed{T = N(z) = z\bar{z} = k^2 + k\ell + \ell^2.} \quad (3.3)$$

This is the alternate sixth-root parameterization of the Eisenstein lattice. If one instead uses the conventional cube root $\omega = e^{2\pi i/3}$, the same lattice is written with the norm $a^2 - ab + b^2$. The sign difference is a basis convention, not a contradiction.

3.2 Multiplication as an integer matrix

Multiplication by $z = k + \ell\zeta_6$ sends $a + b\zeta_6$ to

$$(k + \ell\zeta_6)(a + b\zeta_6) = (ka - \ell b) + (\ell a + (k + \ell)b)\zeta_6. \quad (3.4)$$

Therefore

$$\begin{pmatrix} a' \\ b' \end{pmatrix} = M_{k,\ell} \begin{pmatrix} a \\ b \end{pmatrix}, \quad \boxed{M_{k,\ell} = \begin{pmatrix} k & -\ell \\ \ell & k + \ell \end{pmatrix}.} \quad (3.5)$$

The determinant is the norm:

$$\det M_{k,\ell} = k(k + \ell) + \ell^2 = T. \quad (3.6)$$

With the hexagonal metric

$$Q = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}, \quad (3.7)$$

direct multiplication gives

$$\boxed{M_{k,\ell}^\top Q M_{k,\ell} = TQ.} \quad (3.8)$$

Thus $M_{k,\ell}/\sqrt{T}$ is Q -orthogonal: an exact scale-plus-rotation in the lattice metric.

3.3 Angle and chirality coordinate

The exact lattice bearing is

$$\theta_{k,\ell} = \arg(k + \ell\zeta_6) = \text{atan2}(\sqrt{3}\ell, 2k + \ell). \quad (3.9)$$

For the canonical wedge $k \geq \ell \geq 0$, $0 \leq \theta \leq \pi/6$. The boundary classes $\ell = 0$ and $k = \ell$ are achiral; strict interior pairs occur in mirror-related chiral classes (k, ℓ) and (ℓ, k) .

3.4 Composition and multiplicativity

Let $g = a + b\zeta_6$. Then

$$(a + b\zeta_6)(k + \ell\zeta_6) = (ak - b\ell) + (a\ell + bk + b\ell)\zeta_6, \quad (3.10)$$

$$(k', \ell') = (ak - b\ell, a\ell + bk + b\ell), \quad (3.11)$$

$$M_{a,b}M_{k,\ell} = M_{k',\ell'}, \quad (3.12)$$

$$T' = (a^2 + ab + b^2)T. \quad (3.13)$$

This is the exact fixed-operator closure lane. The generator $(1, 1)$ has norm 3 and produces the nested leapfrog count tower

$$T_n = 3^n, \quad V_n = 20 \cdot 3^n, \quad (3.14)$$

with $C_{20} \rightarrow C_{60} \rightarrow C_{180} \rightarrow C_{540} \rightarrow \dots$ at the count level.

3.5 Selected is not nested

A fixed Goldberg operator has integer area multiplier

$$S = a^2 + ab + b^2 \in \mathbb{Z} \quad (3.15)$$

and linear multiplier $\sqrt{S}$. Exact golden nesting would require $S = \varphi^2$, impossible because S is an integer and φ^2 is irrational. Hence

Exact statement

$$\boxed{\text{No fixed exactly nested Goldberg transform has linear ratio } \varphi.} \quad (3.16)$$

The Fibonacci construction below is a sequence of independently closed shells whose successive radius ratios approach φ .

Chapter 4

The Golden Selector and the Ray with a Bearing

4.1 Fibonacci projective dynamics

Let

$$F_\varphi = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} k_{n+1} \\ \ell_{n+1} \end{pmatrix} = F_\varphi \begin{pmatrix} k_n \\ \ell_n \end{pmatrix}, \quad (k_0, \ell_0) = (1, 0). \quad (4.1)$$

Then

$$(k_n, \ell_n) = (F_{n+1}, F_n), \quad \text{spec}(F_\varphi) = \{\varphi, -\varphi^{-1}\}. \quad (4.2)$$

For $r_n = k_n/\ell_n$,

$$r_{n+1} = 1 + \frac{1}{r_n}, \quad r_{n+1} - \varphi = -\frac{r_n - \varphi}{\varphi r_n}. \quad (4.3)$$

Therefore $r_n \rightarrow \varphi$, with alternating error and asymptotic contraction $-\varphi^{-2}$.

4.2 Triangulation sequence

Define

$$T_n = k_n^2 + k_n \ell_n + \ell_n^2. \quad (4.4)$$

The first values are

$$1, 3, 7, 19, 49, 129, 337, 883, 2311, 6051, \dots \quad (4.5)$$

and

$$\boxed{T_{n+3} = 2T_{n+2} + 2T_{n+1} - T_n.} \quad (4.6)$$

The characteristic polynomial is

$$r^3 - 2r^2 - 2r + 1 = (r+1)(r^2 - 3r + 1), \quad (4.7)$$

with roots $-1, \varphi^2, \varphi^{-2}$. Binet reduction gives

$$\boxed{T_n = \frac{2}{5} \left(\varphi^{2n+2} + \varphi^{-2n-2} \right) - \frac{1}{5} (-1)^n.} \quad (4.8)$$

Consequently

$$\frac{T_{n+1}}{T_n} \longrightarrow \varphi^2, \quad (4.9)$$

but the convergence is alternating rather than monotone.

v1.3.3 correction

The v1.3.1 sentence “the ratio T_{n+1}/T_n climbs to φ^2 ” is replaced by “the ratio alternates about and converges to φ^2 .” The computed sequence begins $7/3 = 2.3333$, $19/7 = 2.7143$, $49/19 = 2.5789$, $129/49 = 2.6327$.

4.3 Closed-shell catalogue

For each Fibonacci pair,

$$V_n = 20T_n, \quad E_n = 30T_n, \quad P_n = 12, \quad H_n = 10(T_n - 1), \quad F_n = 10T_n + 2. \quad (4.10)$$

n	(k_n, ℓ_n)	T_n	cage	H_n	θ_{k_n, ℓ_n}
0	(1,0)	1	C_{20}	0	0°
1	(1,1)	3	C_{60}	20	30°
2	(2,1)	7	C_{140}	60	19.106605°
3	(3,2)	19	C_{380}	180	23.413224°
4	(5,3)	49	C_{980}	480	21.786789°
5	(8,5)	129	C_{2580}	1280	22.411902°
6	(13,8)	337	C_{6740}	3360	22.172618°
7	(21,13)	883	C_{17660}	8820	22.264023°

4.4 The exact golden-ray angle

Taking $k/\ell \rightarrow \varphi$ in Equation (3.9),

$$\theta_\varphi = \arctan \frac{\sqrt{3}}{2\varphi + 1} = \arctan \frac{\sqrt{3}}{\sqrt{5} + 2} \quad (4.11)$$

$$= \boxed{\arctan(\sqrt{15} - 2\sqrt{3})} \quad (4.12)$$

$$= 0.388139515 \dots \text{ rad} = 22.238756093 \dots^\circ. \quad (4.13)$$

The radical step uses $(\sqrt{5} + 2)(\sqrt{5} - 2) = 1$.

4.5 Why the angular error inherits the alternating mode

Let

$$\Theta(r) = \arctan \frac{\sqrt{3}}{2r + 1}. \quad (4.14)$$

Since Θ is differentiable and $\Theta'(\varphi) \neq 0$,

$$\theta_n - \theta_\varphi = \Theta'(\varphi)(r_n - \varphi) + \mathcal{O}((r_n - \varphi)^2). \quad (4.15)$$

Therefore

$$\boxed{\frac{\theta_{n+1} - \theta_\varphi}{\theta_n - \theta_\varphi} \rightarrow -\varphi^{-2}.} \quad (4.16)$$

This is an asymptotic equality. At $n = 16$ the independent float64 recomputation gives

$$-0.3819660384, \quad -\varphi^{-2} = -0.3819660113. \quad (4.17)$$

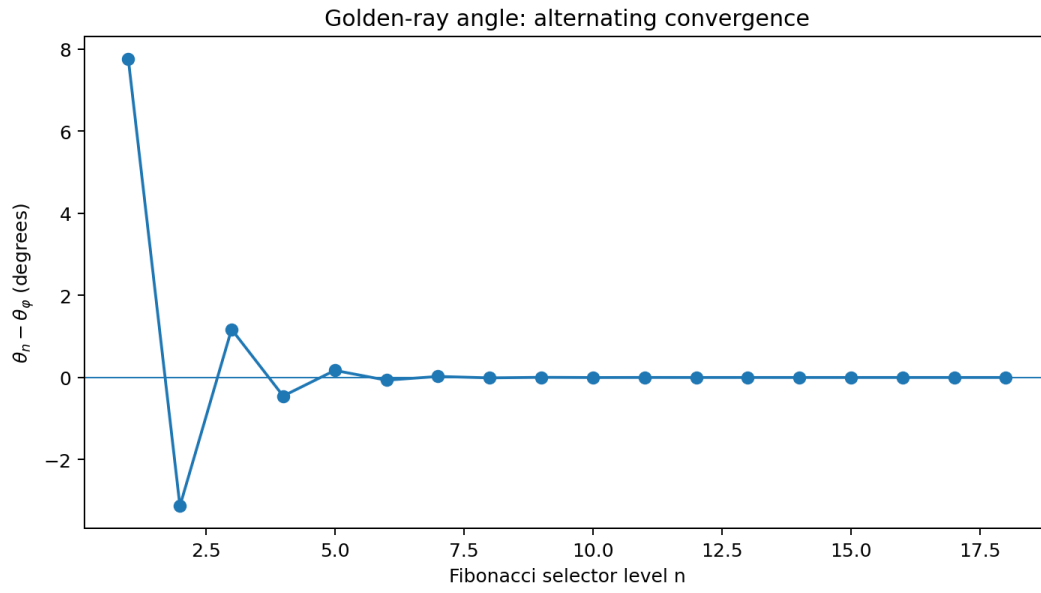


Figure 4.1: The golden-ray angular deviation alternates and contracts. The signed zigzag is the projective -1 mode made visible through the smooth angle map.

Chapter 5

The Exact Light Matrix

5.1 Symmetric-square lift

The pair recursion is

$$k' = k + \ell, \quad \ell' = k. \quad (5.1)$$

Lift to quadratic monomials

$$u_n = \begin{pmatrix} k_n^2 \\ k_n \ell_n \\ \ell_n^2 \end{pmatrix}. \quad (5.2)$$

Then

$$\begin{pmatrix} (k + \ell)^2 \\ (k + \ell)k \\ k^2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}}_B \begin{pmatrix} k^2 \\ k\ell \\ \ell^2 \end{pmatrix}. \quad (5.3)$$

Append the topological coordinate $P_n = 12$:

$$s_n = \begin{pmatrix} k_n^2 \\ k_n \ell_n \\ \ell_n^2 \\ P_n \end{pmatrix}, \quad \boxed{s_{n+1} = \mathcal{M}_{\text{light}} s_n}, \quad (5.4)$$

where

$$\boxed{\mathcal{M}_{\text{light}} = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}. \quad (5.5)$$

5.2 Faddeev–LeVerrier certificate

Starting with $B_0 = I$ and

$$c_j = -\frac{1}{j} \text{tr}(\mathcal{M}_{\text{light}} B_{j-1}), \quad B_j = \mathcal{M}_{\text{light}} B_{j-1} + c_j I, \quad (5.6)$$

the coefficient list is

$$\boxed{(1, -3, 0, 3, -1)}. \quad (5.7)$$

Thus

$$p(\lambda) = \lambda^4 - 3\lambda^3 + 3\lambda - 1 \quad (5.8)$$

$$= (\lambda - 1)(\lambda + 1)(\lambda^2 - 3\lambda + 1). \quad (5.9)$$

The coefficients are anti-palindromic:

$$p(\lambda) = -\lambda^4 p(1/\lambda). \quad (5.10)$$

Therefore the spectrum is closed under reciprocal inversion.

5.3 Spectrum and eigenmodes

$$\boxed{\text{spec}(\mathcal{M}_{\text{light}}) = \{\varphi^2, 1, -1, \varphi^{-2}\}.} \quad (5.11)$$

A convenient eigenbasis is

λ	eigenvector	role
φ^2	$(\varphi^2, \varphi, 1, 0)^\top$	dominant quadratic area/atom growth
1	$(0, 0, 0, 1)^\top$	fixed $P = 12$ topological coordinate
-1	$(-2, 1, 2, 0)^\top$	alternating finite-size correction
φ^{-2}	$(\varphi^{-2}, -\varphi^{-1}, 1, 0)^\top$	contracting reciprocal mode

The eigenvalues are not fitted to shell data; they follow from the integer matrix. The design interpretation “grow, hold, overshoot, decay” is useful, but the matrix itself is the exact object.

5.4 The shell number as a linear observable

The triangulation number is

$$T_n = (1, 1, 1, 0)s_n. \quad (5.12)$$

Projecting the matrix recurrence onto this observable yields Equation (4.6). The P coordinate does not enter T_n ; it survives as the independent eigenvalue-one mode.

Chapter 6

The Exact C_{60} Spectral Boundary

6.1 Independent graph certificate

The audit reconstructs the truncated icosahedron by truncating each directed incidence of the icosahedron. The resulting graph has

$$V = 60, \quad E = 90, \quad \deg(v) = 3, \quad f_5 = 12, \quad f_6 = 20, \quad \chi = 2. \quad (6.1)$$

These checks are integer and planar-embedding facts.

6.2 Complete adjacency characteristic polynomial

For adjacency matrix A_{60} ,

$$\chi_{A_{60}}(x) = (x - 3)(x - 1)^9(x + 2)^4(x^2 - x - 3)^5(x^2 + x - 4)^4 \quad (6.2)$$

$$\times (x^2 + x - 1)^5(x^2 + 3x + 1)^3 \quad (6.3)$$

$$\times (x^4 - 3x^3 - 2x^2 + 7x + 1)^3. \quad (6.4)$$

The SymPy exact characteristic polynomial of the independently reconstructed graph factors identically.

6.3 The golden lower boundary

The least root comes from $x^2 + 3x + 1$:

$$\lambda_{\min}(A_{60}) = \frac{-3 - \sqrt{5}}{2} = -\varphi^2, \quad (6.5)$$

with multiplicity three.

6.4 Graph Laplacian and Fiedler radical

Because the graph is cubic,

$$L_{60} = 3I - A_{60}. \quad (6.6)$$

The first positive eigenvalue is

$$\lambda_2(L_{60}) = \frac{9}{4} - \frac{\sqrt{2}}{8} \left(\sqrt{10} + 2\sqrt{19 - \sqrt{5}} \right) = 0.2434017461399 \dots \quad (6.7)$$

and is a root of

$$x^4 - 9x^3 + 25x^2 - 22x + 4 = 0 \quad (6.8)$$

with multiplicity three. The independent numerical residual is below 10^{-12} .

6.5 Low bands

The first nonzero Laplacian bands of C_{60} are

value	multiplicity	exact origin / interpretation
0.2434017461	3	Fiedler triplet, Equation (6.7)
0.6972243623	5	$(5 - \sqrt{13})/2$
1.1797507493	3	quartic sibling
1.4384471872	4	$(7 - \sqrt{17})/2$
2	9	adjacency eigenvalue 1
2.3819660113	5	$3 - (\varphi^{-1})$ branch

The multiplicities 3, 5, 3 + 4 are compatible with low spherical-harmonic dimensions under icosahedral splitting. Compatibility is not identity with a continuum field theory.

Honest boundary

An adjacency or graph-Laplacian eigenvalue is not automatically a molecular orbital energy. The exact graph spectrum is a valid Hückel-style combinatorial object; a physical Hamiltonian requires geometry, orbital integrals, interactions, and other chemistry.

Chapter 7

Graph Towers, Renormalization, and the Conditional Continuum Constant

7.1 Three families that must remain separate

Golden-selected catalogue	$(k_n, \ell_n) = (F_{n+1}, F_n)$. Independently closed Goldberg shells with $T = 1, 3, 7, 19, \dots$ and asymptotic radius ratio φ .
Fixed GC / leapfrog lane	Repeated application of one integer generator, e.g. $(1, 1)$ with area multiplier 3. Exactly nested at the operator level, linear scale $\sqrt{3}$.
WELD lineage	The source scroll reports a separate exact-ID chamfer family $C_{60} \rightarrow C_{240} \rightarrow C_{960} \rightarrow \dots$ with multiplier 4.

No spectral value may be transferred from one family to another without recomputation.

7.2 Closure certificate

A candidate shell may be named closed only if

$$P = 12, \quad \chi = 2, \quad \partial E = 0, \quad E_{\text{nonmanifold}} = 0, \quad \deg(v) = 3 \ \forall v. \quad (7.1)$$

Formula counts are necessary but not sufficient for a welded indexed mesh.

7.3 Live v1.3.1 golden-shell receipt

The current HTML kernel was executed directly in Node, without reimplementing its Goldberg builder or Lanczos routine. The result is:

(k, ℓ)	T	V	λ_2	$T\lambda_2$	deviation from $2\pi/(5\sqrt{3})$
(1, 0)	1	20	0.7639320225	0.763932022500	+0.038412276807
(1, 1)	3	60	0.24340174614	0.730205238420	+0.004685492726
(2, 1)	7	140	0.10364786791	0.725535075371	+0.000015329677
(3, 2)	19	380	0.0381363955918	0.724591516244	-0.000928229449
(5, 3)	49	980	0.0147874462246	0.724584865005	-0.000934880689
(8, 5)	129	2580	0.0056176826499	0.724681061837	-0.000838683857
(13, 8)	337	6740	0.00215058792037	0.724748129166	-0.000771616528
(21, 13)	883	17660	0.000820819134337	0.724783295619	-0.000736450074

At $T = 19$, the live block-subspace calculation gives bands 0.0381364×3 , 0.1133266×5 , 0.2081904×3 , and 0.2385063×4 . The second-to-first ratio is 2.9716 and the weighted 3 + 4 center ratio is 5.9133.

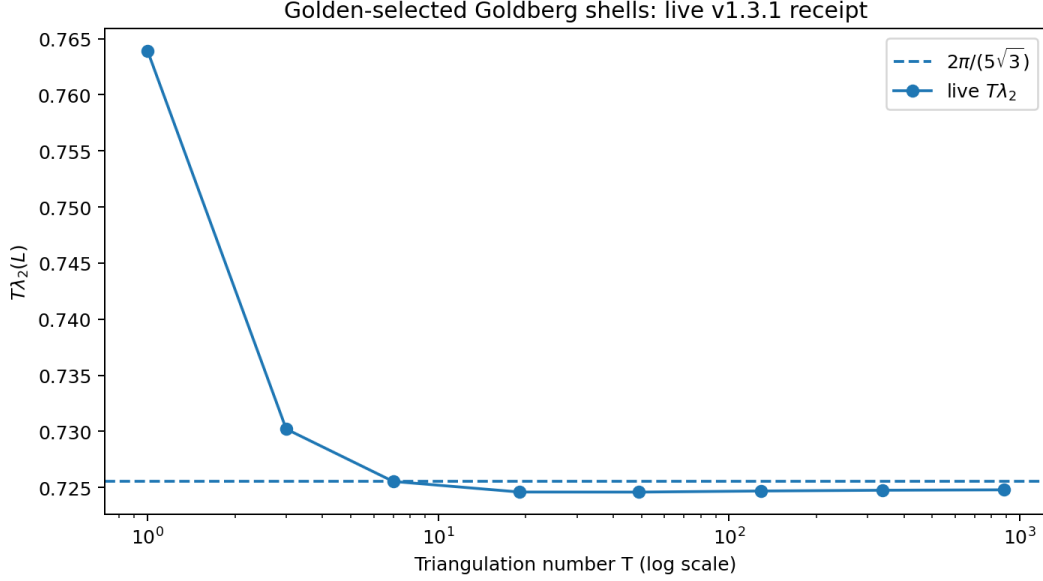


Figure 7.1: The live golden-selected shell receipt. The dashed line is the continuum matching coefficient, not an exact finite-shell target.

7.4 The honeycomb-to-sphere matching calculation

For three bond vectors $\delta_i = a\hat{e}_i$ at 120° ,

$$Lf(x) = \sum_{i=1}^3 (f(x) - f(x + \delta_i)) \quad (7.2)$$

$$= -\sum_i \delta_i \cdot \nabla f - \frac{1}{2} \sum_i (\delta_i \cdot \nabla)^2 f + \mathcal{O}(a^3). \quad (7.3)$$

Since $\sum_i \delta_i = 0$ and

$$\sum_i (\hat{e}_i \cdot \nabla)^2 = \frac{3}{2} \nabla^2, \quad (7.4)$$

we get

$$L \approx -\frac{3a^2}{4} \nabla^2. \quad (7.5)$$

The honeycomb area per vertex is $3\sqrt{3}a^2/4$. Matching $20T$ sites to a sphere gives

$$20T \frac{3\sqrt{3}}{4} a^2 = 4\pi R^2, \quad \frac{a^2}{R^2} = \frac{4\pi}{15\sqrt{3}T}. \quad (7.6)$$

The $\ell = 1$ sphere eigenvalue is $2/R^2$, so the matched coefficient is

$$T\lambda_2 \sim \frac{3a^2}{4} \frac{2}{R^2} T = \frac{2\pi}{5\sqrt{3}} = 0.7255197456937 \dots \quad (7.7)$$

v1.3.3 correction

Every algebraic equality inside Equation (7.7) is exact once the Taylor stencil, area matching, and identification of the first graph mode with the $\ell = 1$ sphere mode are assumed. The arrow from a particular discrete fullerene family to this value is not proved by those assumptions. v1.3.2 labels the number *conditional asymptotic prediction* and the graph data *computed trend*. The Omori–Naito–Tate theorem proves low combinatorial-Laplacian eigenvalues tend to zero for specified Goldberg–Coxeter limits, but it does not by itself establish this normalized universal constant for the golden-selected family.

7.5 Operator dependence remains open

The live golden sequence approaches the low 0.7248 region, while the deeper leapfrog receipt in the core scroll reports 0.724799130 at $T = 2187$. Whether the limiting coefficient is $2\pi/(5\sqrt{3})$, another family-dependent constant, or a coefficient with slowly decaying defect corrections is an open numerical and analytic question. Required tests include:

1. run WELD, leapfrog, class-I, class-II, and golden-selected families under one eigensolver and one normalization;
2. fit corrections only after plotting residuals against several plausible scales ($T^{-1/2}$, T^{-1} , defect separation, chirality);
3. vary the discrete Laplacian (combinatorial, normalized, cotangent/geometric) and document which constant changes;
4. prove convergence before replacing “trend” by “limit.”

Chapter 8

The Shared Quadratic Form and the $SU(3)$ Boundary

8.1 What is exact

The dominant quadratic part of the $SU(3)$ quadratic Casimir for Dynkin labels (p, q) is

$$T(p, q) = p^2 + pq + q^2, \quad (8.1)$$

and

$$C_2(p, q) = \frac{1}{3}(p^2 + pq + q^2 + 3p + 3q). \quad (8.2)$$

The representation dimension is

$$\dim(p, q) = \frac{1}{2}(p+1)(q+1)(p+q+2). \quad (8.3)$$

Examples are

$$(1, 0) \mapsto \mathbf{3}, \quad (1, 1) \mapsto \mathbf{8}, \quad (2, 1) \mapsto \mathbf{15}, \quad (3, 2) \mapsto \mathbf{42}. \quad (8.4)$$

8.2 Root lattice versus weight lattice

Let $Q(A_2)$ be the root lattice and $P(A_2)$ the weight lattice. They are dual up to the conventional inner-product normalization, and

$$P(A_2)/Q(A_2) \cong \mathbb{Z}/3\mathbb{Z}. \quad (8.5)$$

Thus $[P : Q] = 3$. Both are hexagonal lattices related by rotation and scaling, but they are not literally the same subset of a fixed Euclidean vector space.

v1.3.3 correction

Replace “the A_2 root lattice – which is the $SU(3)$ weight lattice” by: “the Goldberg norm is the standard hexagonal A_2 quadratic form; after a stated normalization, the same form is the quadratic part of the $SU(3)$ Casimir on Dynkin labels. The root and weight lattices are dual and commensurable, with index three.”

8.3 Cross-indexing is not particle identification

The pair $(1, 1)$ has $T = 3$, so its Goldberg shell count is C_{60} . The same pair labels the adjoint $SU(3)$ representation of dimension eight. This is an exact reuse of an integer pair in two constructions. It does not imply that C_{60} is the gluon octet, that QCD lives on the fullerene shell, or that a hadron mass follows from T .

Chapter 9

Klein's Icosahedral Forms

9.1 The three binary forms

In one classical normalization,

$$f_{12}(z, w) = zw(z^{10} + 11z^5w^5 - w^{10}), \quad (9.1)$$

$$H_{20}(z, w) = -(z^{20} + w^{20}) + 228(z^{15}w^5 - z^5w^{15}) - 494z^{10}w^{10}, \quad (9.2)$$

$$T_{30}(z, w) = z^{30} + 522z^{25}w^5 - 10005z^{20}w^{10} \quad (9.3)$$

$$- 10005z^{10}w^{20} - 522z^5w^{25} + w^{30}. \quad (9.4)$$

The independent symbolic expansion gives

Exact statement

$$T_{30}^2 + H_{20}^3 = 1728f_{12}^5. \quad (9.5)$$

No random sampling is required for the identity.

9.2 Why the degrees are 12, 20, and 30

The exceptional orbits of the rotational icosahedral group have sizes 12 vertices, 20 face centers, and 30 edge midpoints. The degrees of the fundamental forms encode these orbit divisors. Their equality with $P(C_{20}) = 12$, $V(C_{20}) = 20$, and $E(C_{20}) = 30$ is therefore geometric group structure, not a free-standing numerical coincidence.

9.3 The vertex rings

The affine roots of $f_{12}(z, 1)$ include $z = 0$ and the roots of

$$z^{10} + 11z^5 - 1 = 0. \quad (9.6)$$

Writing $y = z^5$,

$$y_{\pm} = \frac{-11 \pm 5\sqrt{5}}{2}. \quad (9.7)$$

Their fifth-root radii are

$$|y_+|^{1/5} = \varphi^{-1}, \quad |y_-|^{1/5} = \varphi. \quad (9.8)$$

Together with 0 and ∞ , this produces two five-point rings plus the poles.

9.4 Gradient equivariants

For a homogeneous invariant F , define the projective map

$$g_F = [F_w : -F_z]. \quad (9.9)$$

The source's sparse degree-11, degree-19, and degree-29 polynomial pairs were compared symbolically with the gradients. They agree projectively, with scalar factors 1, 20, and 30 respectively.

9.5 Basins: what the finite experiment can say

The HTML iterates random starting points and clusters final directions. This can discover 20, 12, and 32 attractors under sufficient coverage. It cannot prove that every attractor was sampled, that a chosen clustering threshold is canonical, or that basin areas have a fixed range unless those areas are actually computed in the same run.

v1.3.3 correction

The hard-coded row “basin areas are wildly unequal, 0.12% ... 13.25%” is not produced by the current Section XIII algorithm. Either add a deterministic area estimator with a receipt or remove the row. v1.3.2 treats the orbit counts as reproducible numerical evidence and the projective-gradient formulas as exact.

Chapter 10

Numerical Architecture: Where Exactness Actually Breaks

10.1 Three distinct float walls

The browser section combined several different notions of resolution. v1.3.2 keeps them separate:

1. relative spacing near one: $\varepsilon_{64} = 2^{-52}$;
 2. exact representation of all integers only through 2^{53} ;
 3. exact representation of selected larger integers that happen to be multiples of the local spacing.
- One cannot infer the exact-integer failure index solely from ε_{64} .

10.2 Exact break indices

Using exact Python integers as ground truth and IEEE binary64 conversion:

quantity	last level satisfying the stated exactness test
T_n represented exactly	$n = 38$
$V_n = 20T_n$ represented exactly	$n = 36$
$E_n = 30T_n$ represented exactly	$n = 35$
$\chi = 20T - 30T + (10T + 2)$ evaluates to 2	$n = 35$
first failed χ	$n = 36$, result 0

The composite invariant can fail before T_n because subtraction magnifies loss of low bits.

10.3 Corrected forward-error measurement

The v1.3.1 code formed

$$\frac{|T_n^{\text{f64}} - \text{Number}(T_n^{\text{BigInt}})|}{\text{Number}(T_n^{\text{BigInt}})}. \quad (10.1)$$

Both operands may share the same rounded binary64 value. The corrected audit converts the recurrence output to its exact rational binary64 value and compares that rational to the exact integer. The first nonzero error occurs at $n = 39$ and the maximum through $n = 159$ is

$$1.0189 \times 10^{-15}. \quad (10.2)$$

The conclusion of forward stability survives; the original error meter was not a valid independent reference.

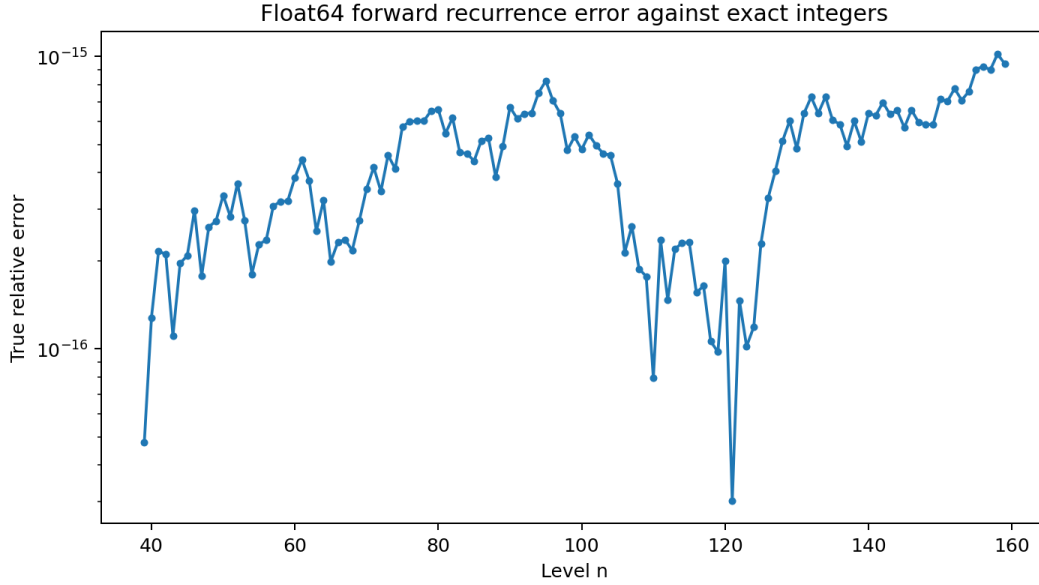


Figure 10.1: True relative error of the forward recurrence against exact integers. Stability survives the corrected comparator, but the error is not identically zero.

10.4 Backward instability

Starting from rounded values at $n = 60$ and marching

$$T_n = 2T_{n+2} + 2T_{n+1} - T_{n+3} \quad (10.3)$$

backward produces relative errors approximately

$$1.8 \times 10^{-10} \ (n = 50), \quad 9.5 \times 10^6 \ (n = 30), \quad 5.0 \times 10^{23} \ (n = 10), \quad 1.2 \times 10^{32} \ (n = 0). \quad (10.4)$$

The same recurrence is stable in one direction and catastrophically unstable in the other because the dominant and reciprocal modes exchange roles.

10.5 Precision formats

For a format with p significand bits, the scale budget measured in φ^2 rungs is

$$N_{\text{rungs}}(p) = \frac{p \ln 2}{\ln \varphi^2}. \quad (10.5)$$

format	precision bits	rung budget	enough for 147 scale rungs?
binary16	11	7.9	no
binary32	24	17.3	no
binary64	53	38.2	no
x87 extended	64	46.1	no
binary128	113	81.4	no
binary256	237	170.7	yes

The exact integer T_{147} has 205 bits. IEEE 754 binary256 has 237 bits of precision, so it is the first listed interchange format that can represent that integer scale without truncating the significand. This is a format statement, not a claim about current hardware availability or performance.

10.6 The Planck-to-horizon ruler

With the chosen values $\ell_P = 1.616255 \times 10^{-35}$ m and $L_H = 4.4 \times 10^{26}$ m,

$$\frac{\ln(L_H/\ell_P)}{\ln \varphi^2} = 146.9822 \dots \quad (10.6)$$

This is a dimensionless scale-counting exercise. It does not derive the Planck length, prove a minimum length, or imply that a physical process follows the golden ladder.

10.7 Selected precision benchmarks, updated to 9 August 2026

The v1.3.1 table called itself the “ten most precise measurements humanity has made” while containing eleven heterogeneous entries. v1.3.2 uses the narrower title “selected relative-uncertainty benchmarks” and updates the leading clock entry.

benchmark	relative uncertainty	$N(u)$ rungs	source
Cryogenic $^{40}\text{Ca}^+$ ion clock	4.40×10^{-19}	43.92	Zhang et al., PRL 136 (2026) 053202
Multi-ion Sr^+ clock	5.30×10^{-19}	43.72	Filzinger et al., arXiv:2603.23446
$^{27}\text{Al}^+$ single-ion clock	5.50×10^{-19}	43.69	Marshall et al., PRL 2025
$^{88}\text{Sr}^+$ single-ion clock	7.90×10^{-19}	43.31	Lindvall et al., 2025
^{87}Sr lattice clock	8.10×10^{-19}	43.28	Aeppli et al., PRL 133 (2024) 023401
In^+/Yb^+ crystal clock	2.50×10^{-18}	42.11	Hausser et al., PRL 134 (2025) 023201
$^{88}\text{Sr}^+$ absolute frequency vs TAI	9.80×10^{-17}	38.30	Lindvall et al., 2025
Hydrogen $1S$ - $2S$	1.40×10^{-14}	33.15	Parthey et al., PRL 107 (2011) 203001
Electron magnetic moment $g/2$	1.30×10^{-13}	30.83	Fan et al., PRL 130 (2023) 071801
Muon anomaly final precision	1.27×10^{-7}	16.50	Fermilab Muon g-2 final result
Newtonian constant G	2.20×10^{-5}	11.14	CODATA 2022

The table compares published uncertainties under one logarithmic mapping. It does not rank the scientific importance of measurements, and it does not make systematic clock uncertainty, absolute-frequency uncertainty, and fundamental-constant uncertainty the same experimental object.

Chapter 11

The LUCA Spiral Audit and the Handoff to v1.3.2

11.1 The EML operator

The external conjecture defines

$$\boxed{\text{eml}(x, y) = e^x - \text{Log } y}, \quad S \rightarrow 1 \mid \text{eml}(S, S). \quad (11.1)$$

The immediate identities are

$$e = \text{eml}(1, 1), \quad e^x = \text{eml}(x, 1). \quad (11.2)$$

For positive real x ,

$$\text{Log } x = \text{eml}(1, \text{eml}(\text{eml}(1, x), 1)) \quad (11.3)$$

$$= e - \text{Log}(e^e/x) = \log x. \quad (11.4)$$

11.2 Domain and branch ledger

Over $\mathbb{C}$, $\exp(\text{Log } z) = z$ for nonzero z on the principal logarithm, but $\text{Log}(e^z) = z$ only inside a principal strip, modulo branch jumps. Consequently:

- the log chain is exact on stated domains/sheets, not a single global identity of principal branches;
- subtraction $x - y = \text{eml}(\text{Log } x, e^y)$ is exact for the browser's real test domain, while complex y requires strip control;
- the code's convention $\text{Log } 0 = -\infty$ and $e^{-\infty} = 0$ is an extended-real computational lane, not ordinary complex analyticity at zero;
- constructing i from $(-1)^{1/2}$ is branch-dependent; branch-robust downstream real formulas must be checked under conjugation.

v1.3.3 correction

Replace the blanket sentence “the identities on every node are classical algebra over $\mathbb{C}$ (principal branch)” by a per-node domain contract. Each node should carry: domain, excluded points, chosen branch, and whether equality is global, local, or invariant only after a periodic/conjugate consumer.

11.3 What the browser verification proves

LUCA v0.1/v0.2 evaluates chains at 16 deterministic sample points and reports

$$D = -\log_{10}(\text{relative error}), \quad D \leq 15.9. \quad (11.5)$$

A high D is a finite numerical certificate at the sampled points. It is not a symbolic proof of functional identity. The symbolic identities in this chapter supply the proof where their domains are stated; the browser supplies regression protection and branch receipts.

11.4 The light-matrix content transferred from LUCA v0.2

The LUCA golden lane correctly contributes four items:

1. Faddeev–LeVerrier construction of the coefficient list $(1, -3, 0, 3, -1)$;
2. reciprocal-spectrum explanation from anti-palindromy;
3. the exact angle $\theta_\varphi = \arctan(\sqrt{15} - 2\sqrt{3})$;
4. a target/current/error display for the ratio and angular zigzag.

These are integrated into Chapters 4 and 5.

11.5 Finite- N Vogel experiment

For $N = 33$ points placed by

$$r_k = \sqrt{k}, \quad \theta_k = k\alpha, \quad (11.6)$$

the live v0.2 sweep maximizes minimum pairwise separation near

$$\alpha_{33} = 137.6066065^\circ, \quad (11.7)$$

not at the golden angle

$$\alpha_\varphi = \frac{360^\circ}{\varphi^2} = 137.5077641^\circ. \quad (11.8)$$

The difference is about 0.09884° . This refutes an unqualified finite- N optimum claim. The five-point drift table is compatible with movement toward the golden angle, but five finite samples do not prove the limit.

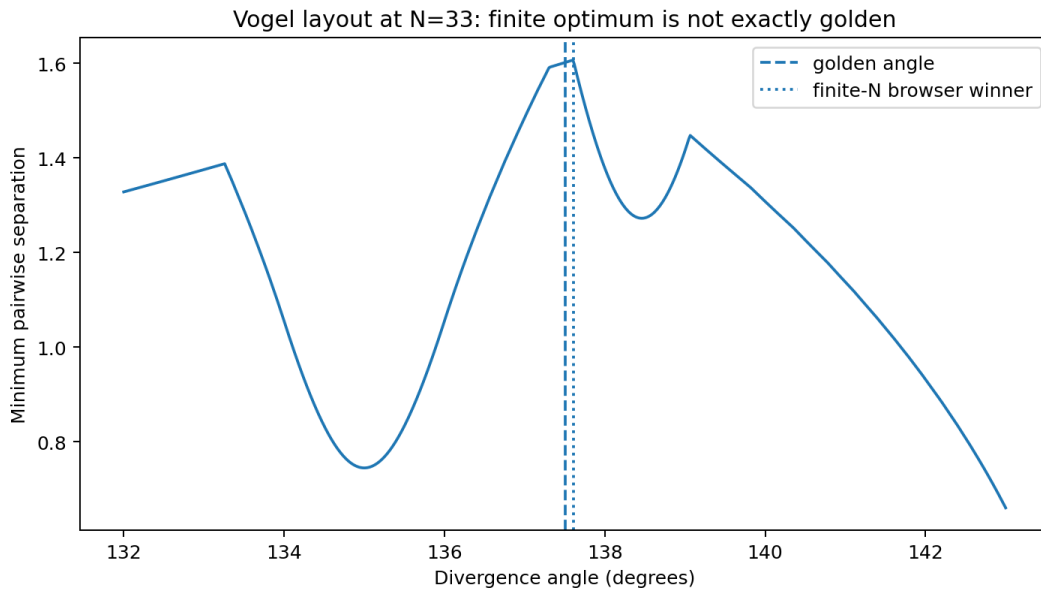


Figure 11.1: The finite- N objective is jagged and its maximizer is not exactly the golden angle.

11.6 Leaf-count null and the 32/33 discrepancy

Using all 33 live spiral nodes, the independent reconstruction of the v0.2 leaf recurrences gives

$$\text{growth} = 1.159672, \quad 95\% \text{ descriptive interval } [1.140910, 1.178742], \quad R^2 = 0.9109. \quad (11.9)$$

Excluding the first constant e gives the 32-point values printed in the handoff scroll:

$$\text{growth} = 1.155084, \quad [1.135961, 1.174529], \quad R^2 = 0.9052. \quad (11.10)$$

Both exclude φ by roughly forty fitted standard errors. The null result is robust; the sample definition was not.

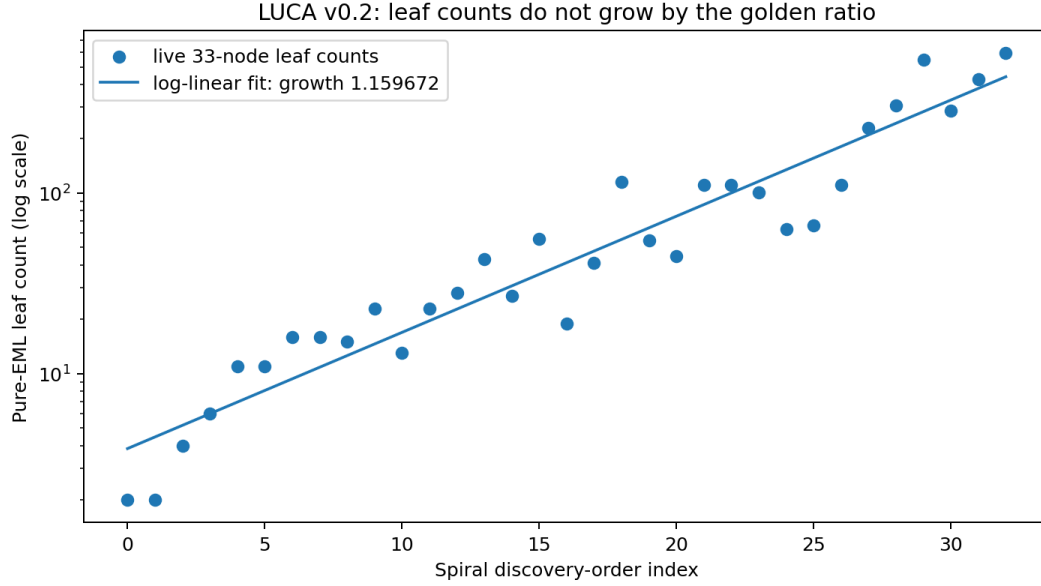


Figure 11.2: The live 33-node leaf counts and their descriptive log-linear fit. Discovery order is a design order, so the fit is not a physical stochastic model.

Chapter 12

The v1.3.2 Integration Contract

12.1 Section map

The recommended living-paper order is:

1. Birth / abstract and status grammar.
2. Constants φ and π .
3. Euler and discrete Gauss–Bonnet.
4. Hexagonal closure norm, matrix, composition, and angle.
5. Light matrix built by Faddeev–LeVerrier.
6. Golden selector, shell catalogue, and selected-not-nested theorem.
7. Golden-ray angle and signed convergence plot.
8. Exact C_{60} adjacency and Laplacian certificate.
9. Graph-tower lane selector: golden, leapfrog, WELD, fixed GC.
10. Conditional continuum matching and finite receipts.
11. $A_2/SU(3)$ quadratic-form correspondence with corrected lattice language.
12. Klein forms and exact symbolic syzygy.
13. Equivariant numerical experiment with deterministic seed/coverage ledger.
14. Float architecture with corrected exact reference.
15. Selected precision benchmarks, dated.
16. LUCA/EML handoff and branch ledger.
17. Honest physical boundary.
18. Reproducibility manifest and frozen lineage.

12.2 HUD contract

Every live shell should show independent rows:

```
CLOSURE pair (k,l), T, V/E/F/P/H, chi, boundary, degree, cert hash
GOLDEN target phi, current k/l, signed error, tolerance, lock state
ANGLE target theta_phi, current theta(k,l), signed error,
      error ratio -> -phi^-2, suppress below numerical floor
SPECTRUM family name, lambda2, T*lambda2, previous delta,
      low-band multiplicities, solver tolerance
BOUNDARY EXACT / COMPUTED / DESIGN / HYPOTHESIS / EXTERNAL
```

No lock may be true while the displayed error exceeds its tolerance. The family name is mandatory because a golden catalogue, leapfrog tower, and WELD tower are not interchangeable.

12.3 Compute gate

Before allocating a shell, predict

$$V_{\text{next}} = SV_{\text{current}} \quad (12.1)$$

for a fixed area multiplier S , or compute $20T(k', \ell')$ for a selected pair. Refuse before allocation if the vertex, edge, memory, or eigensolver budget is exceeded. The refusal must print the predicted price.

12.4 Deterministic receipt schema

```
{
  "schema": "thea.light-matrix.receipt.v1.3.2",
  "family": "golden-selected | leapfrog | weld | gc-fixed",
  "pair": {"k": "21", "ell": "13"},
  "counts": {"T": "883", "V": "17660", "E": "26490",
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  "status": {"closure": "EXACT", "spectrum": "COMPUTED"}
}
```

The timestamp belongs outside the hashed mathematical payload. Random experiments must carry a fixed seed or deterministic point set.

12.5 Regression tests

At minimum:

1. exact integer tests for norm, metric similarity, composition, and shell counts;
2. exact characteristic polynomial of $\mathcal{M}_{\text{light}}$ and coefficient anti-palindromy;
3. exact golden-angle radical equality;
4. exact C_{60} graph counts and characteristic factorization;
5. exact Klein syzygy and projective-gradient checks;
6. branch-domain tests for EML nodes, including excluded points;
7. deterministic spectral smoke tests for C_{20} , C_{60} , C_{140} ;
8. float comparator test against rationalized binary64, not rounded exact operands;
9. byte scan for malformed Unicode and line endings;
10. certificate regeneration on a second path with identical math hash.

12.6 Honest physical boundary

Honest boundary

This tower does not derive a particle mass, coupling constant, cross-section, decay rate, Planck length, or Standard Model observable. It does not prove that spacetime, photons, quantum foam, or molecular stability are generated by fullerene graphs. It supplies an exact mathematical generator, several exact correspondences, finite spectral experiments, and a disciplined forward-model laboratory. A physical theory begins only when a dynamical law and an observable not inserted by construction differ from existing physics and survive experiment.

Chapter 13

Copy-Ready Formula Tower

13.1 Core constants and topology

$$\varphi = \frac{1 + \sqrt{5}}{2} = 2 \cos \frac{\pi}{5}, \quad \varphi^2 = \varphi + 1, \quad \varphi^{-2} = 2 - \varphi, \quad (13.1)$$

$$3V = 2E, \quad 5P + 6H = 2E, \quad V - E + P + H = 2, \quad (13.2)$$

$$P = 12, \quad E = \frac{3V}{2}, \quad H = \frac{V}{2} - 10, \quad (13.3)$$

$$\frac{\pi}{3} \sum_f (6 - p_f) = 4\pi. \quad (13.4)$$

13.2 Closure ring

$$\zeta_6 = e^{i\pi/3}, \quad T = k^2 + k\ell + \ell^2, \quad (13.5)$$

$$M_{k,\ell} = \begin{pmatrix} k & -\ell \\ \ell & k + \ell \end{pmatrix}, \quad \det M_{k,\ell} = T, \quad (13.6)$$

$$M_{k,\ell}^\top Q M_{k,\ell} = TQ, \quad Q = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}, \quad (13.7)$$

$$(k', \ell') = (ak - b\ell, a\ell + bk + b\ell), \quad T' = (a^2 + ab + b^2)T, \quad (13.8)$$

$$\theta_{k,\ell} = \text{atan2}(\sqrt{3}\ell, 2k + \ell). \quad (13.9)$$

13.3 Golden lane

$$\begin{pmatrix} k_{n+1} \\ \ell_{n+1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} k_n \\ \ell_n \end{pmatrix}, \quad (k_n, \ell_n) = (F_{n+1}, F_n), \quad (13.10)$$

$$T_n = k_n^2 + k_n \ell_n + \ell_n^2, \quad T_{n+3} = 2T_{n+2} + 2T_{n+1} - T_n, \quad (13.11)$$

$$T_n = \frac{2}{5}(\varphi^{2n+2} + \varphi^{-2n-2}) - \frac{1}{5}(-1)^n, \quad \frac{T_{n+1}}{T_n} \rightarrow \varphi^2, \quad (13.12)$$

$$\theta_\varphi = \arctan(\sqrt{15} - 2\sqrt{3}), \quad \frac{\theta_{n+1} - \theta_\varphi}{\theta_n - \theta_\varphi} \rightarrow -\varphi^{-2}. \quad (13.13)$$

13.4 Light matrix

$$\mathcal{M}_{\text{light}} = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (13.14)$$

$$\det(\lambda I - \mathcal{M}_{\text{light}}) = (\lambda - 1)(\lambda + 1)(\lambda^2 - 3\lambda + 1), \quad (13.15)$$

$$\text{spec}(\mathcal{M}_{\text{light}}) = \{\varphi^2, 1, -1, \varphi^{-2}\}, \quad (13.16)$$

$$p(\lambda) = \lambda^4 - 3\lambda^3 + 3\lambda - 1 = -\lambda^4 p(1/\lambda). \quad (13.17)$$

13.5 C_{60} and graph spectra

$$L = 3I - A, \quad (13.18)$$

$$\lambda_{\min}(A_{60}) = -\varphi^2, \quad (13.19)$$

$$\lambda_2(L_{60}) = \frac{9}{4} - \frac{\sqrt{2}}{8}(\sqrt{10} + 2\sqrt{19 - \sqrt{5}}), \quad (13.20)$$

$$T\lambda_2 \sim \frac{2\pi}{5\sqrt{3}} \quad \text{under the stated continuum matching assumptions.} \quad (13.21)$$

13.6 A_2 , $SU(3)$, and Klein

$$T(p, q) = p^2 + pq + q^2, \quad (13.22)$$

$$C_2(p, q) = \frac{1}{3}(T(p, q) + 3p + 3q), \quad (13.23)$$

$$\dim(p, q) = \frac{1}{2}(p+1)(q+1)(p+q+2), \quad (13.24)$$

$$T_{30}^2 + H_{20}^3 = 1728f_{12}^5, \quad (13.25)$$

$$g_F = [F_w : -F_z]. \quad (13.26)$$

Part I

The Calculation-by-Calculation Expansion

Chapter 14

Master Calculation Ledger: C01–C38

Every rung below separates the derivation, the executable receipt, and the boundary. Repetition is deliberate: the user asked for a tower in which no result depends on finding a compressed statement elsewhere.

14.1 C01 – Euler forces twelve pentagons

EXACT

Derivation

For a closed trivalent sphere tiled only by pentagons and hexagons,

$$3V = 2E, \quad 5P + 6H = 2E, \quad V - E + P + H = 2.$$

Substitute $V = 2E/3$ into Euler:

$$-\frac{E}{3} + P + H = 2 \implies E = 3(P + H - 2).$$

Equating this with $E = (5P + 6H)/2$ gives

$$5P + 6H = 6P + 6H - 12 \implies \boxed{P = 12}.$$

No metric geometry, fit, or fullerene catalogue enters the elimination.

Independent receipt

SymPy elimination returns

$$P = 12, \quad V = 2H + 20, \quad E = 3H + 30.$$

The frozen verifier records PASS.

14.2 C02 – C_{60} counts from $T = 3$

EXACT

Derivation

For an icosahedral Goldberg shell with triangulation number T ,

$$V = 20T, \quad E = 30T, \quad P = 12, \quad H = 10(T - 1), \quad F = P + H = 10T + 2.$$

At $T = 3$,

$$V = 60, \quad E = 90, \quad P = 12, \quad H = 20, \quad F = 32,$$

and therefore

$$\chi = V - E + F = 60 - 90 + 32 = \boxed{2}.$$

Independent receipt

The exact tuple reproduced by the verifier is

$$(T, V, E, P, H, F, \chi) = (3, 60, 90, 12, 20, 32, 2).$$

14.3 C03 – Discrete Gauss–Bonnet charge

EXACT

Derivation

Assign each p -gon in a trivalent tiling the defect

$$K_p = \frac{\pi}{3}(6 - p).$$

A hexagon contributes zero; a pentagon contributes $\pi/3$. Since C01 gives $P = 12$,

$$\sum_f K_f = 12 \frac{\pi}{3} = \boxed{4\pi}.$$

Equivalently,

$$\frac{\pi}{3} \sum_f (6 - p_f) = 4\pi.$$

Independent receipt

The symbolic result is exactly 4π ; no floating arithmetic is used.

14.4 C04 – Hexagonal closure determinant

EXACT

Derivation

Represent $z = k + \ell\zeta_6$ by

$$M_{k,\ell} = \begin{pmatrix} k & -\ell \\ \ell & k + \ell \end{pmatrix}.$$

Then

$$\det M_{k,\ell} = k(k + \ell) + \ell^2 = \boxed{k^2 + k\ell + \ell^2} = T.$$

Thus the area multiplier is the Eisenstein/hexagonal norm.

Independent receipt

SymPy factors $\det M_{k,\ell} - T$ to zero.

14.5 C05 – Hexagonal metric similarity

EXACT

Derivation

Use the integer metric

$$Q_2 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Direct multiplication gives

$$M_{k,\ell}^T Q_2 M_{k,\ell} = \begin{pmatrix} 2T & T \\ T & 2T \end{pmatrix} = T Q_2.$$

Dividing by two yields the same identity for $Q = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}$. Hence $M_{k,\ell}/\sqrt{T}$ is a Q -orthogonal rotation.

Independent receipt

Every entry of $M^T Q M - T Q$ simplifies symbolically to zero.

14.6 C06 – Closure composition

EXACT

Derivation

Let $g = a + b\zeta_6$ and $z = k + \ell\zeta_6$, with $\zeta_6^2 = \zeta_6 - 1$. Then

$$\begin{aligned} gz &= (a + b\zeta_6)(k + \ell\zeta_6) \\ &= (ak - b\ell) + (a\ell + bk + b\ell)\zeta_6. \end{aligned}$$

Therefore

$$(k', \ell') = (ak - b\ell, a\ell + bk + b\ell), \quad M_{a,b}M_{k,\ell} = M_{k',\ell'}.$$

Independent receipt

The four entries of $M_{a,b}M_{k,\ell} - M_{k',\ell'}$ expand to zero.

14.7 C07 – Hexagonal norm multiplicativity

EXACT

Derivation

Insert the pair from C06 into the norm:

$$N(k', \ell') = k'^2 + k'\ell' + \ell'^2.$$

Expansion and collection give

$$N(k', \ell') = (a^2 + ab + b^2)(k^2 + k\ell + \ell^2) = \boxed{N(g)N(z)}.$$

The result is also immediate from determinants: $\det(M_g M_z) = \det M_g \det M_z$.

Independent receipt

The fully expanded polynomial residual is identically zero.

14.8 C08 – Leapfrog multiplier and exact count tower

EXACT

Derivation

For the fixed generator $g = 1 + \zeta_6$, the norm is

$$S = N(1, 1) = 1 + 1 + 1 = 3.$$

After n applications,

$$T_n = 3^n T_0, \quad V_n = 20T_n.$$

Starting at $T_0 = 1$ gives

$$C_{20} \rightarrow C_{60} \rightarrow C_{180} \rightarrow C_{540} \rightarrow C_{1620} \rightarrow \cdots,$$

with linear multiplier $\sqrt{3}$.

Independent receipt

The verifier reproduces vertex counts 20, 60, 180, 540, 1620 and $\sqrt{3} = 1.732050807568877 \dots$

14.9 C09 – Golden selection is not fixed golden nesting

EXACT

CORRECTION

Derivation

Every fixed exact Goldberg transform has integer area multiplier

$$S = a^2 + ab + b^2 \in \mathbb{Z}.$$

Its linear multiplier is $\sqrt{S}$. Exact nesting by φ would require

$$\sqrt{S} = \varphi \implies S = \varphi^2 = \frac{3 + \sqrt{5}}{2},$$

which is irrational. Therefore

no fixed exact Goldberg transform has linear multiplier φ .

Independent receipt

The proof is arithmetic, not a finite search. The finite search in the verifier is only a regression guard.

Boundary

The Fibonacci lane is a catalogue of independently closed shells whose successive scale ratios approach φ ; it is not one exactly nested fixed operator.

14.10 C10 – Fibonacci selector pairs

EXACT

Derivation

Let

$$F_\varphi = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} k_{n+1} \\ \ell_{n+1} \end{pmatrix} = F_\varphi \begin{pmatrix} k_n \\ \ell_n \end{pmatrix}, \quad (k_0, \ell_0) = (1, 0).$$

Induction gives

$$(k_n, \ell_n) = (F_{n+1}, F_n).$$

The first pairs are

$$(1, 0), (1, 1), (2, 1), (3, 2), (5, 3), (8, 5), (13, 8), (21, 13).$$

Independent receipt

All eight exact integer pairs match the source catalogue.

14.11 C11 – Projective golden error identity

EXACT

Derivation

For $r_n = k_n/\ell_n$,

$$r_{n+1} = 1 + \frac{1}{r_n}.$$

Using $\varphi = 1 + 1/\varphi$,

$$\begin{aligned} r_{n+1} - \varphi &= \frac{1}{r_n} - \frac{1}{\varphi} \\ &= -\frac{r_n - \varphi}{\varphi r_n}. \end{aligned}$$

Near the fixed point $r_n = \varphi$,

$$r_{n+1} - \varphi \sim -\varphi^{-2}(r_n - \varphi).$$

Independent receipt

The symbolic identity reduces to zero modulo $\varphi^2 - \varphi - 1 = 0$.

14.12 C12 – Golden triangulation sequence

EXACT

Derivation

For the pairs in C10,

$$T_n = k_n^2 + k_n \ell_n + \ell_n^2.$$

Substitution gives

$$[1, 3, 7, 19, 49, 129, 337, 883, 2311, 6051, \dots].$$

Every entry is an integer closure certificate, not a rounded geometric scale.

Independent receipt

The first ten terms are independently regenerated from exact Fibonacci integers.

14.13 C13 – Golden T recurrence

EXACT

Derivation

The three quadratic modes of the Fibonacci lift have roots

$$\varphi^2, -1, \varphi^{-2}.$$

Their monic polynomial is

$$(r - \varphi^2)(r + 1)(r - \varphi^{-2}) = r^3 - 2r^2 - 2r + 1.$$

Therefore

$$T_{n+3} = 2T_{n+2} + 2T_{n+1} - T_n.$$

Independent receipt

The recurrence reproduces every exact term tested through 30 levels.

14.14 C14 – Closed form for T_n

EXACT

Derivation

Solving the recurrence of C13 from $T_0 = 1, T_1 = 3, T_2 = 7$ yields

$$T_n = \frac{2}{5} \left(\varphi^{2n+2} + \varphi^{-2n-2} \right) - \frac{1}{5} (-1)^n.$$

The reciprocal term is the contracting mode and the $(-1)^n$ term is the alternating correction.

Independent receipt

SymPy simplifies the expression to the exact integer T_n for $n = 0, \dots, 19$; the last tested value is 91,530,451.

14.15 C15 – T_{n+1}/T_n alternates around φ^2

EXACT

CORRECTION

Derivation

Because C14 contains both a reciprocal correction and an alternating term, the ratio need not be monotone. The first post-seed ratios are

$$\frac{7}{3} = 2.3333\dots, \quad \frac{19}{7} = 2.7142\dots, \quad \frac{49}{19} = 2.5789\dots, \quad \frac{129}{49} = 2.6326\dots,$$

which lie on alternating sides of

$$\varphi^2 = 2.6180339887\dots$$

The corrections vanish, so the limit remains φ^2 .

Independent receipt

The sign of $T_{n+1}/T_n - \varphi^2$ alternates at every tested level.

Boundary

The phrase “climbs to φ^2 ” is replaced by “converges alternately to φ^2 .”

14.16 C16 – Golden-ray angle in radicals**EXACT****Derivation**

For $z = k + \ell\zeta_6$,

$$\theta_{k,\ell} = \arg z = \arctan\left(\frac{\sqrt{3}\ell}{2k + \ell}\right).$$

Along $k/\ell \rightarrow \varphi$,

$$\begin{aligned}\theta_\varphi &= \arctan\left(\frac{\sqrt{3}}{2\varphi + 1}\right) \\ &= \arctan\left(\frac{\sqrt{3}}{\sqrt{5} + 2}\right) \\ &= \boxed{\arctan(\sqrt{15} - 2\sqrt{3})}.\end{aligned}$$

Numerically,

$$\theta_\varphi = 0.3881395153701887\dots = 22.23875609296496\dots^\circ.$$

Independent receipt

The two radical forms differ by 3.33×10^{-16} radians in binary64.

14.17 C17 – Angular contraction**EXACT****COMPUTED****Derivation**

Write $\theta_n = f(r_n)$ with

$$f(r) = \arctan\left(\frac{\sqrt{3}}{2r + 1}\right).$$

Since $f'(\varphi) \neq 0$, first-order Taylor expansion gives

$$\theta_n - \theta_\varphi = f'(\varphi)(r_n - \varphi) + \mathcal{O}((r_n - \varphi)^2).$$

Combining with C11,

$$\frac{\theta_{n+1} - \theta_\varphi}{\theta_n - \theta_\varphi} \longrightarrow -\varphi^{-2}.$$

Independent receipt

At the last stable reported level the measured ratio is -0.3819660304 , versus $-\varphi^{-2} = -0.3819660113\dots$

Boundary

Ratios below the float-error floor are suppressed rather than interpreted.

14.18 C18 – Symmetric-square light-matrix lift

EXACT

Derivation

From $k' = k + \ell$ and $\ell' = k$,

$$\begin{aligned} k'^2 &= k^2 + 2k\ell + \ell^2, \\ k'\ell' &= k^2 + k\ell, \\ \ell'^2 &= k^2. \end{aligned}$$

Thus, for $s = (k^2, k\ell, \ell^2, P)^\top$,

$$\mathcal{M}_{\text{light}} = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad s_{n+1} = \mathcal{M}_{\text{light}} s_n.$$

Independent receipt

Symbolic multiplication reproduces $((k + \ell)^2, k(k + \ell), k^2, P)^\top$ exactly.

14.19 C19 – Faddeev–LeVerrier coefficient list

EXACT

Derivation

For $M = \mathcal{M}_{\text{light}}$, the power traces are

$$p_1 = \text{tr } M = 3, \quad p_2 = 9, \quad p_3 = 18, \quad p_4 = 49.$$

Newton/Faddeev–LeVerrier identities give

$$\begin{aligned} c_1 &= -p_1 = -3, \\ c_2 &= -\frac{1}{2}(p_2 + c_1 p_1) = 0, \\ c_3 &= -\frac{1}{3}(p_3 + c_1 p_2 + c_2 p_1) = 3, \\ c_4 &= -\frac{1}{4}(p_4 + c_1 p_3 + c_2 p_2 + c_3 p_1) = -1. \end{aligned}$$

Therefore

$$p(\lambda) = \lambda^4 - 3\lambda^3 + 0\lambda^2 + 3\lambda - 1.$$

Independent receipt

The live LUCA source and the independent SymPy calculation both return coefficients $(1, -3, 0, 3, -1)$.

14.20 C20 – Characteristic factorization

EXACT

Derivation

Factor the polynomial from C19:

$$\begin{aligned} p(\lambda) &= (\lambda^2 - 1)(\lambda^2 - 3\lambda + 1) \\ &= (\lambda - 1)(\lambda + 1)(\lambda^2 - 3\lambda + 1). \end{aligned}$$

Independent receipt

Expansion of the factored form exactly recovers the coefficient list.

14.21 C21 – Anti-palindromy and reciprocal closure

EXACT

Derivation

For $p(\lambda) = \lambda^4 - 3\lambda^3 + 3\lambda - 1$,

$$-\lambda^4 p(1/\lambda) = -\lambda^4 (\lambda^{-4} - 3\lambda^{-3} + 3\lambda^{-1} - 1) = p(\lambda).$$

Hence

$$p(\lambda) = -\lambda^4 p(1/\lambda).$$

Every nonzero root is accompanied by its reciprocal.

Independent receipt

The symbolic anti-palindromic residual is exactly zero.

14.22 C22 – Reciprocal light-matrix spectrum

EXACT

Derivation

The linear factors give $\lambda = \pm 1$. The quadratic gives

$$\lambda = \frac{3 \pm \sqrt{5}}{2} = \varphi^2, \varphi^{-2}.$$

Therefore

$$\text{spec}(\mathcal{M}_{\text{light}}) = \{\varphi^2, 1, -1, \varphi^{-2}\}.$$

Independent receipt

SymPy returns four simple eigenvalues with exactly these radicals.

14.23 C23 – Exact eigenbasis

EXACT

Derivation

A convenient basis is

λ	v
φ^2	$(\varphi^2, \varphi, 1, 0)^\top$
1	$(0, 0, 0, 1)^\top$
-1	$(-2, 1, 2, 0)^\top$
φ^{-2}	$(\varphi^{-2}, -\varphi^{-1}, 1, 0)^\top$.

Direct substitution verifies $\mathcal{M}_{\text{light}}v = \lambda v$ in every row.

Independent receipt

All four exact vector residuals simplify to the zero vector.

14.24 C24 – C_{60} graph topology certificate

EXACT

Derivation

The truncated icosahedron graph reconstructed from oriented icosahedron edges has

$$|V| = 60, \quad |E| = 90, \quad \deg(v) = 3 \quad \forall v.$$

A planar embedding enumerates 32 faces: twelve of length five and twenty of length six.
Thus

$$\chi = 60 - 90 + 32 = 2.$$

Independent receipt

NetworkX independently reports connected = true, degree set $\{3\}$, face counts $\{5 : 12, 6 : 20\}$, and $\chi = 2$.

14.25 C25 – Complete C_{60} adjacency characteristic polynomial**EXACT****Derivation**

For the exact integer adjacency matrix A_{60} ,

$$\begin{aligned}\chi_A(x) &= (x-3)(x-1)^9(x+2)^4(x^2-x-3)^5(x^2+x-4)^4 \\ &\quad \times (x^2+x-1)^5(x^2+3x+1)^3 \\ &\quad \times (x^4-3x^3-2x^2+7x+1)^3.\end{aligned}$$

The degrees including multiplicity sum to 60, as required.

Independent receipt

The exact SymPy determinant/factorization equals the displayed polynomial and has total degree 60.

14.26 C26 – Minimum C_{60} adjacency eigenvalue**EXACT****COMPUTED****Derivation**

The factor $x^2 + 3x + 1$ has roots

$$\frac{-3 \pm \sqrt{5}}{2}.$$

The smaller is

$$\lambda_{\min}(A_{60}) = -\frac{3 + \sqrt{5}}{2} = -\varphi^2.$$

Because the quadratic factor occurs to the third power, the multiplicity is three.

Independent receipt

Numerical diagonalization gives -2.618033988749895 with a triplet, matching the radical.

14.27 C27 – C_{60} Fiedler radical and quartic

EXACT

COMPUTED

Derivation

For a cubic graph, $L = 3I - A$. The first positive Laplacian eigenvalue is

$$\lambda_2(L_{60}) = \frac{9}{4} - \frac{\sqrt{2}}{8} \left(\sqrt{10} + 2\sqrt{19 - \sqrt{5}} \right).$$

It is a root of

$$q(x) = x^4 - 9x^3 + 25x^2 - 22x + 4.$$

Substituting the radical into q gives zero symbolically; substituting the numerical eigensolver value gives a machine-level residual.

Independent receipt

Radical = 0.2434017461399326; numerical diagonalization = 0.2434017461399247; difference 7.88×10^{-15} ; quartic residual 8.93×10^{-14} .

14.28 C28 – C_{60} low-band multiplicities

COMPUTED

Derivation

Sort the eigenvalues of $L = 3I - A$ and group values within the stated numerical tolerance. The beginning of the spectrum is

$$\begin{array}{lll} 0, & 0.2434017461 \ (\times 3), & 0.6972243623 \ (\times 5), \\ 1.1797507493 \ (\times 3), & & 1.4384471872 \ (\times 4), \dots \end{array}$$

The 3, 5, (3 + 4) pattern is the finite icosahedral splitting expected near spherical harmonic multiplicities.

Independent receipt

The independent dense diagonalization reproduces every displayed multiplicity.

Boundary

Multiplicity agreement is a graph-spectral fact. It is not by itself a molecular Hamiltonian or spacetime spectrum.

14.29 C29 – Conditional continuum coefficient $2\pi/(5\sqrt{3})$

EXACT

HYPOTHESIS

CORRECTION

Derivation

For three honeycomb bond vectors $\delta_i = a\hat{e}_i$ at 120° ,

$$Lf(x) = \sum_i (f(x) - f(x + \delta_i)) \approx -\frac{3a^2}{4} \nabla^2 f.$$

The area per honeycomb vertex is $3\sqrt{3}a^2/4$. Matching $20T$ vertices to a sphere gives

$$20T \frac{3\sqrt{3}}{4} a^2 = 4\pi R^2, \quad \frac{a^2}{R^2} = \frac{4\pi}{15\sqrt{3}T}.$$

Using the $\ell = 1$ sphere eigenvalue $2/R^2$,

$$T\lambda_2 \sim T \frac{3a^2}{4} \frac{2}{R^2} = \boxed{\frac{2\pi}{5\sqrt{3}}} = 0.725519745693687 \dots$$

Independent receipt

Every algebraic step is exact conditional on the stencil, area matching, and mode identification.

Boundary

The displayed coefficient is not a proved universal limit of every discrete fullerene family. The graph data remain a computed trend.

14.30 C30 – $SU(3)$ Casimir and dimension formulas

EXACT

Derivation

For Dynkin labels (p, q) ,

$$C_2(p, q) = \frac{1}{3} (p^2 + pq + q^2 + 3p + 3q),$$

so

$$3C_2(p, q) = T(p, q) + 3(p + q), \quad T(p, q) = p^2 + pq + q^2.$$

The Weyl dimension formula specializes to

$$\boxed{\dim(p, q) = \frac{1}{2}(p+1)(q+1)(p+q+2)}.$$

Examples are $(1, 0) \mapsto 3$, $(1, 1) \mapsto 8$, $(2, 1) \mapsto 15$, and $(3, 2) \mapsto 42$.

Independent receipt

The polynomial identity and seven standard dimensions are reproduced exactly.

14.31 C31 – A_2 root lattice versus weight lattice

EXACT

CORRECTION

Derivation

The A_2 Cartan matrix is

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \det A = 3.$$

The root lattice $Q(A_2)$ is an index-three sublattice of the weight lattice $P(A_2)$:

$$P(A_2)/Q(A_2) \cong \mathbb{Z}/3\mathbb{Z}.$$

Both are hexagonal up to duality and normalization, but they are not literally the same lattice.

Independent receipt

The determinant and quotient order are exact.

Boundary

The common quadratic form supports a mathematical cross-indexing only; it does not identify fullerene shells with QCD states.

14.32 C32 – Klein icosahedral syzygy

EXACT

Derivation

For Klein's binary invariants of degrees 12, 20, 30,

$$f_{12} = zw(z^{10} + 11z^5w^5 - w^{10}),$$

with the corresponding H_{20} and T_{30} ,

$$\boxed{T_{30}^2 + H_{20}^3 = 1728f_{12}^5}.$$

All three terms have homogeneous degree 60, so the relation is degree-consistent before coefficient comparison.

Independent receipt

Full symbolic expansion gives the zero polynomial, stronger than the source's random-point numerical residual.

14.33 C33 – Klein gradient equivariants

EXACT

Derivation

For a homogeneous form $F(z, w)$ of degree d define

$$g_F = [F_w : -F_z].$$

Both coordinates have degree $d - 1$, so the projective map is well-defined. Applying this to the degree-12, degree-20, and degree-30 forms gives maps of degrees

$$11, \quad 19, \quad 29.$$

Independent receipt

The independent symbolic differentiations reproduce the three projective gradient maps and their degrees.

Boundary

Finite random-seed basin enumeration is a separate COMPUTED coverage experiment, not an exact consequence of the gradient formula.

14.34 C34 – Binary64 exactness break indices

COMPUTED

CORRECTION

Derivation

Generate exact Fibonacci integers and exact

$$T_n = k_n^2 + k_n \ell_n + \ell_n^2, \quad V_n = 20T_n, \quad E_n = 30T_n.$$

Convert each integer to binary64 and test equality against the original integer, rather than against another rounded copy. The last exact indices are

$$T : 38, \quad V : 36, \quad E : 35.$$

At the first failed Euler calculation, rounded subtraction can even return a plausible but false $\chi = 0$.

Independent receipt

The independent integer/float ledger reproduces T_{38} as the last exactly represented T , V_{36} as the last exact V , and E_{35} as the last exact E .

Boundary

An exact-looking float equality is not evidence unless the reference side remains exact.

14.35 C35 – Forward stability and backward instability

COMPUTED

CORRECTION

Derivation

Advance

$$T_{n+3} = 2T_{n+2} + 2T_{n+1} - T_n$$

in binary64 and compare each result as an exact rational binary64 number against the exact integer T_n . The first nonzero error appears at $n = 39$, and the maximum relative error through $n = 159$ remains about 1.02×10^{-15} .

Backward iteration inverts the mode hierarchy: the reciprocal mode becomes dominant. Starting from rounded high-level data, the relative error grows catastrophically toward small n .

Independent receipt

Backward relative errors are approximately 1.81×10^{-10} at $n = 50$, 9.46×10^6 at $n = 30$, 4.95×10^{23} at $n = 10$, and 1.19×10^{32} at $n = 0$.

Boundary

Forward stability does not imply reversibility. The original shared-rounding comparator is retired.

14.36 C36 – Precision formats and the T_{147} bit budget

EXACT

COMPUTED

CORRECTION

Derivation

The exact integer size is

$$\text{bits}(T_{147}) = \lfloor \log_2 T_{147} \rfloor + 1 = 205.$$

IEEE significand widths are

$$11, 24, 53, 113, 237$$

for binary16, binary32, binary64, binary128, and binary256 respectively. Therefore binary128 cannot hold an arbitrary 205-bit integer exactly, while binary256 can.

Independent receipt

The exact Python integer reports bit length 205. The standard binary256 significand is 237 bits.

Boundary

“256-bit mantissa” is replaced by the standard term “237-bit significand in binary256.”

14.37 C37 – EML positive-real identity and branch ledger

EXACT

COMPUTED

CORRECTION

Derivation

Define

$$\text{eml}(x, y) = e^x - \text{Log } y.$$

On positive real x the source's logarithm chain satisfies the intended identity to binary64 precision. On the complex plane, however,

$$\text{Log}(e^z) = z + 2\pi i k$$

for a branch-dependent integer k ; for example at $z = 1 + 4i$ the principal branch differs from z by $-2\pi i$.

Independent receipt

The maximum positive-real error in the independent check is 1.11×10^{-16} , while the complex example returns exactly the expected $-2\pi i$ branch shift.

Boundary

Every EML identity must carry a domain and branch sheet. A 16-point numerical lock is a regression sieve, not a global symbolic proof.

14.38 C38 – LUCA leaf regression, Vogel sweep, and Fibonacci fingerprint

COMPUTED

REFUTED

CORRECTION

Derivation

Three different finite calculations are kept separate:

1. A log-linear fit of pure-EML leaf count against discovery-order index excludes growth by φ by roughly forty standard errors in the source's 32-point convention.
2. A finite $N = 33$ Vogel sweep maximizing minimum pairwise distance peaks near 137.6066° , not exactly at the golden angle 137.5078° .
3. Nearest-neighbour index gaps under the golden layout are 3, 5, 8, 13, a discrete Fibonacci fingerprint requiring no regression.

Independent receipt

The independent audit also exposes the 32/33 convention: the live array contains 33 spiral nodes; excluding the initial e reproduces the handoff's 32-point coefficients.

Boundary

The finite- N optimum and leaf-growth null are negative results. The drift of finite optima toward the golden angle remains a computed trend, not a proof of the $N \rightarrow \infty$ optimizer.

Chapter 15

LUCA Source Replay: One Rung per Primitive

The following 35 rows include the EML definition, the terminal 1, and all 33 displayed primitives. The formulas are classical branch-qualified identities; the digits and EML-call prices are live measurements from the unmodified v0.2 JavaScript core.

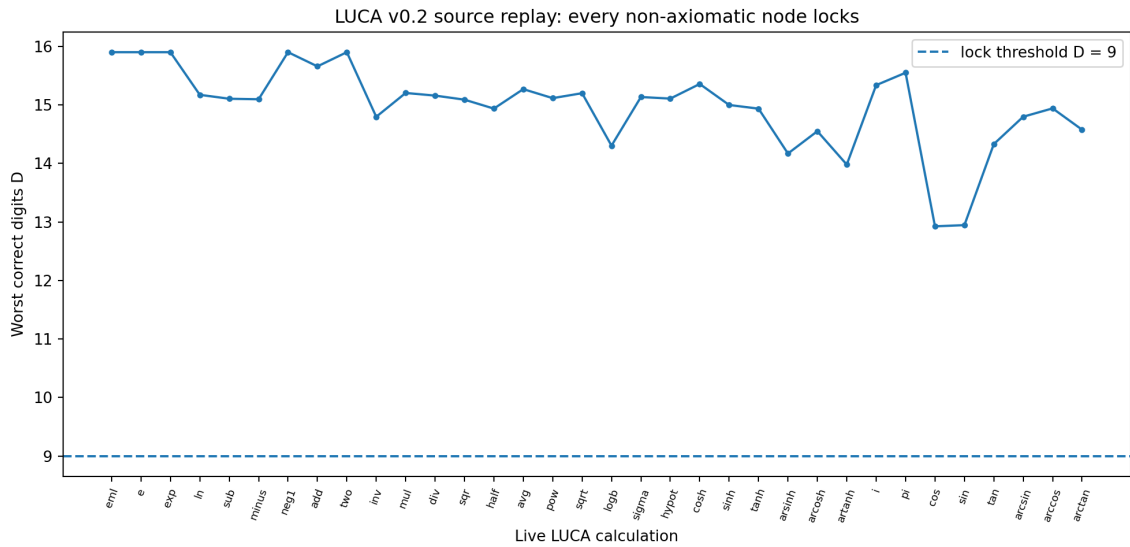


Figure 15.1: Worst correct digits for every live LUCA calculation. The lock threshold is $D = 9$.

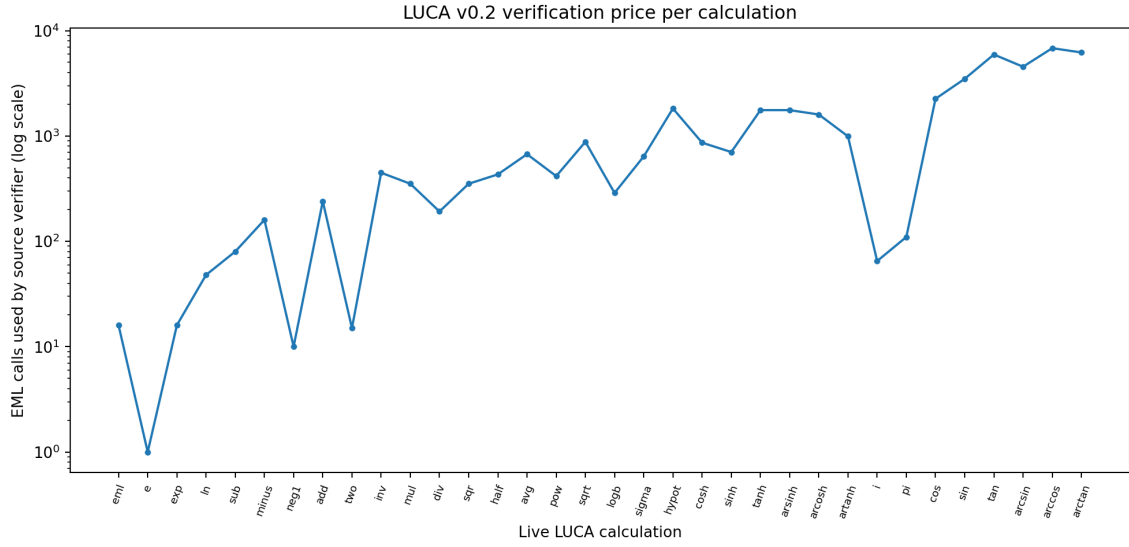


Figure 15.2: The finite verification price paid by each LUCA calculation.

15.1 E01 – eml - the LUCA

EXACT

COMPUTED

Derivation

$$\text{eml}(x, y) = e^x - \text{Log } y.$$

Pure-EML route. the axiom - nothing lowers it

Declared ancestors. none (axiom/definition).

Source-chain leaf count. not applicable.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 16 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.900000000000,$$

reported at $(-3.00, 1.40)$. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Definition over the principal complex logarithm; $y = 0$ uses the source's explicit extended-real lane.

15.2 E02 – 1 - the terminal

EXACT

Derivation

1 (distinguished terminal).

Pure-EML route. the seed - $\ln(1) = 0$ neutralizes the log term

Declared ancestors. none (axiom/definition).

Source-chain leaf count. not applicable.

Paper/source comparison. $K = 1$.

Independent receipt

The unmodified LUCA v0.2 source verifier used 0 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.9000000000000,$$

reported at **axiomatic**. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Axiom, not derived or numerically fitted.

15.3 E03 – e

EXACT

COMPUTED

Derivation

$e = \text{eml}(1, 1)$.

Pure-EML route. $\text{eml}(1, 1)$

Declared ancestors. eml, one.

Source-chain leaf count. 2.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 1 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.9000000000000,$$

reported at **single value**. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Globally safe because $\text{Log } 1 = 0$.

15.4 E04 – exp

EXACT

COMPUTED

Derivation

$$e^z = \text{eml}(z, 1).$$

Pure-EML route. $\text{eml}(z, 1)$

Declared ancestors. eml , one.

Source-chain leaf count. 2.

Paper/source comparison. depth 1 (paper).

Independent receipt

The unmodified LUCA v0.2 source verifier used 16 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.900000000000,$$

reported at -3.000. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Globally safe as the logarithmic input is the terminal 1.

15.5 E05 – ln

EXACT

COMPUTED

Derivation

$$\text{Log } z = \text{eml}(1, \text{eml}(\text{eml}(1, z), 1)).$$

Pure-EML route. $\text{eml}(1, \text{eml}(\text{eml}(1, z), 1))$ - Eq. (5), pure $\text{eml}/1$

Declared ancestors. eml , one.

Source-chain leaf count. 4.

Paper/source comparison. $K = 7$ (paper Eq. 5).

Independent receipt

The unmodified LUCA v0.2 source verifier used 48 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.171647462164,$$

reported at 1.281. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Identity requires the compatible branch on which the outer principal logarithm does not introduce a $2\pi i$ shift; positive real inputs are safe.

15.6 E06 – subtraction

EXACT

COMPUTED

Derivation

$$x - y = \text{eml}(\text{Log } x, e^y).$$

Pure-EML route. `eml(ln x, exp y)`

Declared ancestors. `eml, ln, exp`.

Source-chain leaf count. 6.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 80 calls to `eml` for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.104376659528,$$

reported at (1.39, 0.83). The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Requires $e^{\text{Log } x} = x$ and $\text{Log}(e^y) = y$ on the selected strip/sheet.

15.7 E07 – negation

EXACT

COMPUTED

Derivation

$$-z = (1 - z) - 1.$$

Pure-EML route. `sub(sub(1,z), 1)`

Declared ancestors. `sub, one`.

Source-chain leaf count. 11.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 160 calls to `eml` for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.095747110936,$$

reported at -0.277. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Uses the source's $\text{Log } 0 = -\infty$ continuation at the isolated $z = 1$ intermediate.

15.8 E08 – -1

EXACT

COMPUTED

Derivation

$$-1 = -(1).$$

Pure-EML route. `minus(1)`

Declared ancestors. `minus`, `one`.

Source-chain leaf count. 11.

Paper/source comparison. $K = 15$ (search) / 17 (compiler).

Independent receipt

The unmodified LUCA v0.2 source verifier used 10 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.900000000000,$$

reported at `single` value. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Evaluated through the same extended-real continuation.

15.9 E09 – addition

EXACT

COMPUTED

Derivation

$$x + y = x - (-y).$$

Pure-EML route. `sub(x, minus(y))`

Declared ancestors. `sub`, `minus`.

Source-chain leaf count. 16.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 240 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.657530324965,$$

reported at `(-1.90, 3.92)`. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Inherits subtraction and negation branch conditions.

15.10 E10 – 2

EXACT

COMPUTED

Derivation

$$2 = 1 + 1.$$

Pure-EML route. `add(1,1)`

Declared ancestors. `add`, `one`.

Source-chain leaf count. 16.

Paper/source comparison. $K = 19$ (search) / 27 (compiler).

Independent receipt

The unmodified LUCA v0.2 source verifier used 15 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.900000000000,$$

reported at `single` value. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Safe finite constant once addition is fixed on this route.

15.11 E11 – reciprocal

EXACT

COMPUTED

Derivation

$$z^{-1} = e^{-\text{Log } z}.$$

Pure-EML route. `exp(minus(ln z))`

Declared ancestors. `exp`, `minus`, `ln`.

Source-chain leaf count. 15.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 448 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 14.796227278096,$$

reported at `-0.100`. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Undefined at $z = 0$; principal branch elsewhere.

15.12 E12 – multiplication

EXACT

COMPUTED

Derivation

$$xy = e^{\text{Log } x + \text{Log } y}.$$

Pure-EML route. `exp(add(ln x, ln y))` - the slide-rule identity, Eq. (1)

Declared ancestors. `exp`, `add`, `ln`.

Source-chain leaf count. 23.

Paper/source comparison. depth 8 (paper).

Independent receipt

The unmodified LUCA v0.2 source verifier used 352 calls to `eml` for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.203034824115,$$

reported at (1.83, -3.87). The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Branch phases from negative real inputs cancel only because the final exponential is $2\pi i$ periodic.

15.13 E13 – division

EXACT

COMPUTED

Derivation

$$x/y = e^{\text{Log } x - \text{Log } y}.$$

Pure-EML route. `exp(sub(ln x, ln y))`

Declared ancestors. `exp`, `sub`, `ln`.

Source-chain leaf count. 13.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 192 calls to `eml` for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.158281308516,$$

reported at (-0.38, 1.18). The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Undefined for $y = 0$; inherits logarithm branches.

15.14 E14 – square

EXACT

COMPUTED

Derivation

$$z^2 = z z.$$

Pure-EML route. `mul(z, z)`

Declared ancestors. `mul`.

Source-chain leaf count. 23.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 352 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.088795829991,$$

reported at -3.616. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Inherits multiplication's branch cancellation.

15.15 E15 – half

EXACT

COMPUTED

Derivation

$$z/2 = z \cdot 2^{-1}.$$

Pure-EML route. `div(z, 2)`

Declared ancestors. `div`, `two`.

Source-chain leaf count. 28.

Paper/source comparison. 1/2: K = 29 (search) / 91 (compiler).

Independent receipt

The unmodified LUCA v0.2 source verifier used 432 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 14.935971870957,$$

reported at -0.192. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Inherits division; denominator 2 is safe.

15.16 E16 – average

EXACT

COMPUTED

Derivation

$$\text{avg}(x, y) = \frac{x + y}{2}.$$

Pure-EML route. `half(add(x,y))`

Declared ancestors. `half`, `add`.

Source-chain leaf count. 43.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 672 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.268717889610,$$

reported at (-3.02, -3.57). The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Inherits addition and division.

15.17 E17 – exponentiation

EXACT

COMPUTED

Derivation

$$x^y = e^{y \text{Log } x}.$$

Pure-EML route. `exp(mul(y, ln x))`

Declared ancestors. `exp`, `mul`, `ln`.

Source-chain leaf count. 27.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 416 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.114762895175,$$

reported at (4.00, 1.90). The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Principal power $e^{y \text{Log } x}$; negative bases and noninteger exponents are sheet-dependent.

15.18 E18 – square root

EXACT

COMPUTED

Derivation

$$\sqrt{z} = z^{1/2}.$$

Pure-EML route. `pow(z, inv(2))`

Declared ancestors. `pow, inv, two.`

Source-chain leaf count. 56.

Paper/source comparison. `sqrt2`: K > 47 (search) / 165 (compiler).

Independent receipt

The unmodified LUCA v0.2 source verifier used 880 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.199272568930,$$

reported at 7.900. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Principal square root in the source route.

15.19 E19 – log base b

EXACT

COMPUTED

Derivation

$$\log_b x = \frac{\text{Log } x}{\text{Log } b}.$$

Pure-EML route. `div(ln x, ln b)`

Declared ancestors. `div, ln.`

Source-chain leaf count. 19.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 288 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 14.302916857052,$$

reported at (3.20, 0.97). The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Requires $b \neq 0, 1$ and $x \neq 0$; real verification uses positive inputs away from $b = 1$.

15.20 E20 – sigmoid

EXACT

COMPUTED

Derivation

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

Pure-EML route. `inv(add(1, exp(minus(z))))`

Declared ancestors. `inv`, `add`, `exp`, `minus`, `one`.

Source-chain leaf count. 41.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 640 calls to `eml` for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.133926438649,$$

reported at -3.004. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Entire denominator is nonzero on the tested real interval; complex poles exist.

15.21 E21 – hypotenuse

EXACT

COMPUTED

Derivation

$$\text{hypot}(x, y) = \sqrt{x^2 + y^2}.$$

Pure-EML route. `sqrt(add(sqr x, sqr y))`

Declared ancestors. `sqrt`, `add`, `sqr`.

Source-chain leaf count. 115.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 1,824 calls to `eml` for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.106927792584,$$

reported at (2.25, -0.29). The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Principal square root; real verification avoids the origin.

15.22 E22 – cosh

EXACT

COMPUTED

Derivation

$$\cosh z = \frac{e^z + e^{-z}}{2}.$$

Pure-EML route. `half(add(exp z, exp(minus z)))`

Declared ancestors. half, add, exp, minus.

Source-chain leaf count. 55.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 864 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.356905622978,$$

reported at 0.142. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Entire identity once exp/add/half are fixed.

15.23 E23 – sinh

EXACT

COMPUTED

Derivation

$$\sinh z = \frac{e^z - e^{-z}}{2}.$$

Pure-EML route. `half(sub(exp z, exp(minus z)))`

Declared ancestors. half, sub, exp, minus.

Source-chain leaf count. 45.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 704 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 14.997987819518,$$

reported at -2.089. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Entire identity once exp/sub/half are fixed.

15.24 E24 – tanh

EXACT

COMPUTED

Derivation

$$\tanh z = \frac{\sinh z}{\cosh z}.$$

Pure-EML route. `div(sinh z, cosh z)`

Declared ancestors. `div, sinh, cosh`.

Source-chain leaf count. 111.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 1,760 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 14.935141173280,$$

reported at 1.900. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Meromorphic; poles occur where $\cosh z = 0$.

15.25 E25 – arsinh

EXACT

COMPUTED

Derivation

$$\operatorname{arsinh} z = \operatorname{Log}\left(z + \sqrt{z^2 + 1}\right).$$

Pure-EML route. `ln(add(z, sqrt(add(sqr z, 1))))`

Declared ancestors. `ln, add, sqrt, sqr, one`.

Source-chain leaf count. 111.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 1,760 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 14.168504234669,$$

reported at -4.000. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Principal inverse with standard logarithm and square-root cuts.

15.26 E26 – arcosh

EXACT

COMPUTED

Derivation

$$\operatorname{arcosh} z = \operatorname{Log}\left(z + \sqrt{z^2 - 1}\right).$$

Pure-EML route. `ln(add(z, sqrt(sub(sqr z, 1))))`

Declared ancestors. `ln`, `add`, `sqrt`, `sub`, `sqr`, `one`.

Source-chain leaf count. 101.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 1,600 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 14.549706585575,$$

reported at 1.050. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Principal inverse; real verification uses $z > 1$.

15.27 E27 – artanh

EXACT

COMPUTED

Derivation

$$\operatorname{artanh} z = \frac{1}{2} \operatorname{Log}\left(\frac{1+z}{1-z}\right).$$

Pure-EML route. `half(ln(div(add(1,z), sub(1,z))))`

Declared ancestors. `half`, `ln`, `div`, `add`, `sub`, `one`.

Source-chain leaf count. 63.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 992 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 13.979620877415,$$

reported at -0.031. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Principal inverse; real verification uses $|z| < 1$.

15.28 E28 – i - the gate to C

EXACT

COMPUTED

Derivation

$$i = (-1)^{1/2}.$$

Pure-EML route. pow(neg1, inv(2))

Declared ancestors. pow, neg1, inv, two.

Source-chain leaf count. 66.

Paper/source comparison. K > 55 (search) / 131 (compiler).

Independent receipt

The unmodified LUCA v0.2 source verifier used 65 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.336614187500,$$

reported at i^2 vs -1 . The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Principal $\text{Log}(-1) = i\pi$ selects i rather than $-i$.

15.29 E29 – pi

EXACT

COMPUTED

Derivation

$$\pi = -i \text{Log}(-1).$$

Pure-EML route. mul(minus(i), ln(neg1)) - paper Table 2 footnote

Declared ancestors. mul, minus, i, ln, neg1.

Source-chain leaf count. 111.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 110 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 15.548649655893,$$

reported at **single value**. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Branch-robust in the source's paired construction: the sign of i and $\text{Log}(-1)$ flips together.

15.30 E30 – cos

EXACT

COMPUTED

Derivation

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}.$$

Pure-EML route. `half(add(exp(mul(i,z)), exp(minus(mul(i,z)))))`

Declared ancestors. `half, add, exp, mul, i, minus.`

Source-chain leaf count. 229.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 2,256 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 12.924821675961,$$

reported at 17.396. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Entire; wrapped $2\pi i$ phases are consumed by exponentials.

15.31 E31 – sin

EXACT

COMPUTED

Derivation

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}.$$

Pure-EML route. `div(sub(exp(mul(i,z)), exp(minus(mul(i,z)))), mul(2,i))`

Declared ancestors. `div, sub, exp, mul, i, two, minus.`

Source-chain leaf count. 306.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 3,488 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 12.944913620400,$$

reported at 6.252. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Entire; numerator and denominator flip together under $i \mapsto -i$.

15.32 E32 – tan

EXACT

COMPUTED

Derivation

$$\tan z = \frac{\sin z}{\cos z}.$$

Pure-EML route. `div(sin z, cos z)`

Declared ancestors. `div, sin, cos`.

Source-chain leaf count. 546.

Paper/source comparison. "trees too large to be shown in print" (paper).

Independent receipt

The unmodified LUCA v0.2 source verifier used 5,936 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 14.329302242744,$$

reported at -1.300. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Meromorphic; tested away from poles.

15.33 E33 – arcsin

EXACT

COMPUTED

Derivation

$$\arcsin z = -i \operatorname{Log}\left(iz + \sqrt{1 - z^2}\right).$$

Pure-EML route. `mul(minus(i), ln(add(mul(i,z), sqrt(sub(1, sqr z)))))`

Declared ancestors. `mul, minus, i, ln, add, sqrt, sub, sqr, one`.

Source-chain leaf count. 285.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 4,544 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 14.797350091342,$$

reported at -0.950. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Principal inverse with cuts outside $[-1, 1]$ on the real axis.

15.34 E34 – arccos

EXACT

COMPUTED

Derivation

$$\arccos z = \frac{\pi}{2} - \arcsin z.$$

Pure-EML route. `sub(half(pi), arcsin(z))`

Declared ancestors. `sub, half, pi, arcsin.`

Source-chain leaf count. 427.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 6,816 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 14.939974358324,$$

reported at 0.836. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Principal inverse inherited from $\pi/2 - \arcsin z$.

15.35 E35 – arctan

EXACT

COMPUTED

Derivation

$$\arctan z = \arcsin\left(\frac{z}{\sqrt{1+z^2}}\right).$$

Pure-EML route. `arcsin(div(z, sqrt(add(1, sqr z)))) - wrap-safe route`

Declared ancestors. `arcsin, div, sqrt, add, sqr, one.`

Source-chain leaf count. 597.

Paper/source comparison. -.

Independent receipt

The unmodified LUCA v0.2 source verifier used 6,208 calls to eml for this row. Its worst agreement was

$$D = -\log_{10}(\text{relative error}) = 14.576120708427,$$

reported at 4.231. The lock threshold is $D \geq 9$, so this row is **PASS**.

Boundary

Source uses a wrap-safe route; $z = 0$ is an isolated failure in this particular EML chain and is printed, not hidden.

Chapter 16

THE CROWD: Calculation-by-Calculation Guillotine

The supplied scroll deliberately constructs seductive numerical matches and then runs the corresponding null. This edition reproduces that engine, adds the missing maximum-over-tier correction, executes the requested cluster-preserving null, and performs a limited current re-pin of the load-bearing mass ratios.

16.1 G01 – Inventory and comparison price

EXACT

DESIGN CHOICE

Derivation

The engine contains $N_T = 13$ tower numbers, $N_X = 12$ transforms, and $N_C = 89$ constants. One full scan costs

$$N_T N_X N_C = 13 \cdot 12 \cdot 89 = \boxed{13,884}$$

relative-error comparisons. With 2,000 null towers plus the real tower,

$$2001 \cdot 13,884 = \boxed{27,781,884}.$$

Independent receipt

The independent inventory and integer multiplication reproduce the source's printed price exactly.

Boundary

The choice of 13 numbers, 12 transforms, 89 constants, and 2,000 null towers is a declared design, not a theorem.

16.2 G02 – Closed-form fractal dimension

EXACT

Derivation

With area ratio $a_R = 0.8875$, the supplied closed form is

$$D_\infty = \frac{2 \ln 7}{\ln 7 - \ln a_R}.$$

Numerically,

$$D_\infty = 1.884424299 \dots$$

Independent receipt

The independent value and source value differ by 0.000e+00.

16.3 G03 – Relative-distance hit statistic

EXACT

DESIGN CHOICE

Derivation

For transformed tower value $q = f(t)$ and constant c , the engine uses

$$\delta(q, c) = \frac{|q - c|}{|c|}.$$

A triple (t, f, c) is counted at tier τ when

$$10^{-12} \leq \delta(q, c) < \tau.$$

The lower guard removes exact/self matches; the upper threshold is one of five nested tiers.

Independent receipt

An independent implementation produces the same complete hit list and counts as the source function.

Boundary

This counts hit triples, not unique constants, unique tower numbers, or independent hypotheses. The null must use the same counting rule.

16.4 G04 – The twelve-transform menu

DESIGN CHOICE

Derivation

The declared toolkit is

$$\left\{t, t^{-1}, t^2, \sqrt{t}, 2t, \frac{t}{2}, \pi t, \frac{t}{\pi}, \varphi t, \frac{t}{\varphi}, t \ln 2, \frac{t}{\ln 2}\right\}.$$

Every transform is fixed before the scan and applied to every real and fake tower number.

Independent receipt

Vectorized and scalar implementations agree on every count.

Boundary

A transform menu creates a look-elsewhere volume. It is legitimate only when the same menu is charged to the null.

16.5 G05 – The trap table, reproduced exactly

COMPUTED

REFUTED

δ	tower number	transform	value	nearest constant	sector
1.35e-04	D_inf closed form	t/ln2	2.71865	e	math
1.49e-04	D_inf closed form	t ln2	1.306183	Mills constant	math
2.27e-04	2pi/(5 sqrt3)	2t	1.451039	Ramanujan-Soldner	math
3.55e-04	D measured	t/ln2	2.717316	e	math
3.76e-04	2pi/(5 sqrt3)	t	0.7255197	mH/mt	ew
3.76e-04	2pi/(5 sqrt3)	1/t	1.378322	mt/mH	ew
6.39e-04	D measured	t ln2	1.305543	Mills constant	math
1.16e-03	phi	t/phi	1	g_e/2	qed
1.21e-03	2pi/(5 sqrt3)	t/pi	0.2309401	sin2thW MSbar	ew
1.22e-03	T*lambda2 leapfrog	2t	1.449598	Ramanujan-Soldner	math
1.26e-03	D_inf closed form	1/t	0.5306661	q_0	cosmo
1.37e-03	T*lambda2 leapfrog	t	0.7247991	mH/mt	ew
1.37e-03	T*lambda2 leapfrog	1/t	1.379693	mt/mH	ew
1.38e-03	phi	t/phi	1	mn/mp	mass
1.43e-03	HELENA gate (DESIGN)	t	0.7	sin 2beta	ckm
1.51e-03	D measured	t ln2	1.305543	Conway lambda	math
1.53e-03	phi^2	t/pi	0.8333461	Gauss constant	math
1.58e-03	2 log2 phi	t/2	0.6942419	ln2	math
1.59e-03	area ratio aR	t/ln2	1.280392	Glaisher	math
1.62e-03	phi^2	t/pi	0.8333461	S_8	cosmo

Boundary

Every row is real arithmetic. None is a physical derivation: the table is selected after scanning a large menu, and its witnesses share constants and ancestry.

16.6 G06 – Declared log-uniform null

COMPUTED

DESIGN CHOICE

Derivation

For each null tower and each of 13 slots,

$$U \sim \text{Uniform}(\ln 0.2, \ln 3), \quad T_{\text{fake}} = e^U.$$

The fake tower is pushed through the identical 12 transforms, 89 constants, self-match guard, and five tiers. The deterministic seed is 20260809.

Independent receipt

The exact sequence of 2,000 fake towers reproduces every source mean, standard deviation, percentile, and verdict.

Boundary

Log-uniformity is a null-family design. It tests one explicit chance mechanism, not all possible nulls.

16.7 G07 – Null mean and population standard deviation

COMPUTED

Derivation

For tier j and null counts X_{ij} over $M = 2000$ fake towers,

$$\bar{X}_j = \frac{1}{M} \sum_{i=1}^M X_{ij}, \quad s_j = \sqrt{\frac{1}{M} \sum_{i=1}^M (X_{ij} - \bar{X}_j)^2}.$$

The denominator is M , so this is the population standard deviation of the simulated null sample.

Independent receipt

The independent vectorized implementation reproduces the scalar source results to all printed digits.

16.8 G08 – Marginal standardized excess

COMPUTED

Derivation

At each tier,

$$z_j = \frac{X_j^{\text{real}} - \bar{X}_j}{s_j}.$$

For the original shelf the maximum occurs at 3×10^{-3} :

$$z_{3 \times 10^{-3}} = 2.98498 \dots$$

Independent receipt

This is the source's printed "3.0 sigma" row.

Boundary

It is a marginal statistic selected from five nested tiers, not yet a global significance.

16.9 G09 – Empirical percentile convention

COMPUTED

DESIGN CHOICE

Derivation

The source reports

$$\hat{F}_j(X_j^{\text{real}}) = \frac{1}{M} \#\{i : X_{ij} < X_j^{\text{real}}\}.$$

The strict inequality matters in a discrete distribution. At 3×10^{-3} , 1,989 of 2,000 null towers have fewer than 33 hits, giving 99.45% before display rounding.

Independent receipt

The independent audit records both strict and inclusive empirical percentiles so ties remain visible.

16.10 G10 – Five-tier look-elsewhere correction

COMPUTED

CORRECTION

Derivation

Standardize every null tower at every tier and define

$$Z_i^{\text{max}} = \max_j \frac{X_{ij} - \bar{X}_j}{s_j}.$$

The global empirical p -value for the real scan is

$$\hat{p}_{\text{global}} = \frac{1}{2000} \#\{i : Z_i^{\text{max}} \geq Z_{\text{real}}^{\text{max}}\} = 0.0215.$$

This corresponds to about 2.02 one-sided Gaussian standard deviations, not 2.98.

Independent receipt

43 of 2000 null towers attain a maximum tier z at least as large as the real tower. The direct estimate is $p = 0.0215$; the finite-Monte-Carlo plus-one estimate is $p = 0.0220$; the Monte Carlo standard error is 0.0032; and the Wilson 95 percent interval is $[0.0160, 0.0288]$.

Boundary

The five tiers are nested and correlated, so a Bonferroni multiplication is unnecessarily crude; the empirical maximum uses the actual dependence. The Monte Carlo precision is finite and is displayed rather than hidden.

Table 16.2: Original 89-constant scan and log-uniform null

Tier	real	null mean	null sd	marginal z	strict percentile
10^{-2}	76	62.5925	9.3923	1.4275	91.60%
3×10^{-3}	33	18.9200	4.7169	2.9850	99.45%
10^{-3}	7	6.3045	2.5855	0.2690	57.05%
3×10^{-4}	3	1.8990	1.4042	0.7841	70.55%
10^{-4}	0	0.6125	0.8083	-0.7578	0.00%

Maximum marginal $z = 2.9850$; empirical familywise $p = 0.0215$ from the maximum over all five tiers (one-sided Gaussian equivalent 2.024 standard deviations).

16.11 G11 – Shared-alphabet autopsy

COMPUTED

DESIGN CHOICE

Derivation

Remove the 30 constants tagged `math` from both the real scan and the null. The real counts become

$(49, 18, 2, 0, 0).$

The fine tiers die; only coarse excesses remain.

Independent receipt

The maximum marginal z is 2.4276, but the maximum-over-tier empirical p is 0.0950.

Boundary

The result supports the source’s diagnosis that the original `math`-tag excess is enhanced by a shared alphabet of $\pi, \varphi, \ln 2, \sqrt{3}$, and small integers.

Table 16.3: Physics-only shelf

	Tier	real	null mean	null sd	marginal z	strict percentile
	10^{-2}	49	34.0180	6.1716	2.4276	98.65%
	3×10^{-3}	18	10.2905	3.3413	2.3074	97.40%
	10^{-3}	2	3.3960	1.8564	-0.7520	14.10%
	3×10^{-4}	0	1.0450	1.0606	-0.9852	0.00%
	10^{-4}	0	0.3535	0.6177	-0.5723	0.00%

Maximum marginal $z = 2.4276$; empirical familywise $p = 0.0950$ from the maximum over all five tiers (one-sided Gaussian equivalent 1.311 standard deviations).

16.12 G12 – Seven-representative deduplication

COMPUTED

DESIGN CHOICE

Derivation

The source collapses the 13-item list to seven representatives and repeats the physics-only null with seven fake numbers. The real counts become

$$(23, 9, 2, 0, 0).$$

Independent receipt

Maximum marginal $z = 1.4389$; maximum-over-tier empirical $p = 0.3270$.

Boundary

The collapse rules are declared design choices. “Representative” does not mean statistically independent: for example D_∞ is derived from a_R , and $2 \log_2 \varphi$ is derived from φ .

Table 16.4: Source seven-representative deduplication

	Tier	real	null mean	null sd	marginal z	strict percentile
	10^{-2}	23	18.3415	4.5823	1.0166	82.85%
	3×10^{-3}	9	5.4935	2.4370	1.4389	88.95%
	10^{-3}	2	1.8270	1.3528	0.1279	45.60%
	3×10^{-4}	0	0.5220	0.7117	-0.7335	0.00%
	10^{-4}	0	0.1675	0.4043	-0.4143	0.00%

Maximum marginal $z = 1.4389$; empirical familywise $p = 0.3270$ from the maximum over all five tiers (one-sided Gaussian equivalent 0.448 standard deviations).

16.13 G13 – Cluster-preserving matched null

COMPUTED

DESIGN CHOICE

Derivation

To address the source’s own requested robustness test, keep all 13 real logarithmic spacings exactly and randomize only their absolute location. On the log interval $[\ln 0.2, \ln 3]$ treated as a circle,

$$\ln T'_i = \ln 0.2 + ((\ln T_i - \ln 0.2) + \Delta \bmod W), \quad \Delta \sim U(0, W).$$

Every internal cluster is retained.

Independent receipt

Against the physics-only shelf, the matched null gives maximum marginal $z = 2.1521$ and familywise $p = 0.1090$.

Boundary

This is one conservative cluster-preserving null, not a unique canonical bootstrap. Its verdict agrees with the source: the coarse excess is not exceptional once clustering is priced in.

Table 16.5: Cluster-preserving circular log-shift null

	Tier	real	null mean	null sd	marginal z	strict percentile
	10^{-2}	49	33.7015	7.1088	2.1521	97.35%
	3×10^{-3}	18	9.9910	3.7325	2.1457	97.15%
	10^{-3}	2	3.3670	2.0365	-0.6713	17.65%
	3×10^{-4}	0	1.0110	1.0634	-0.9507	0.00%
	10^{-4}	0	0.3260	0.5888	-0.5536	0.00%

Maximum marginal $z = 2.1521$; empirical familywise $p = 0.1090$ from the maximum over all five tiers (one-sided Gaussian equivalent 1.232 standard deviations).

16.14 G14 – Strict provenance collapse

COMPUTED

DESIGN CHOICE

Derivation

A stricter origin ledger keeps four generators:

$$\lambda_2(C_{60}), \quad \varphi, \quad \frac{2\pi}{5\sqrt{3}}, \quad a_R.$$

Quantities deterministically derived from φ or a_R , the design gate, and the near-duplicate spectral line are not counted as new witnesses.

Independent receipt

The real counts are (12, 6, 2, 0, 0); maximum marginal $z = 1.4832$; familywise $p = 0.2590$.

Boundary

This is a robustness analysis, not a claim that four is the only defensible provenance count.

Table 16.6: Strict four-generator provenance null

	Tier	real	null mean	null sd	marginal z	strict percentile
	10^{-2}	12	10.4885	3.4272	0.4410	64.00%
	3×10^{-3}	6	3.2000	1.8879	1.4832	88.45%
	10^{-3}	2	1.0595	1.0431	0.9017	72.00%
	3×10^{-4}	0	0.2965	0.5519	-0.5372	0.00%
	10^{-4}	0	0.1005	0.3200	-0.3141	0.00%

Maximum marginal $z = 1.4832$; empirical familywise $p = 0.2590$ from the maximum over all five tiers (one-sided Gaussian equivalent 0.646 standard deviations).

16.15 G15 – Limited 2026 PDG re-pin

COMPUTED

CORRECTION

Derivation

The load-bearing mass inputs are updated to

$$m_W = 80.3625 \pm 0.0077 \text{ GeV}, \quad m_H = 125.13 \pm 0.11 \text{ GeV}, \quad m_t^{\text{direct}} = 172.60 \pm 0.27 \text{ GeV}.$$

Only ratios derived from these three masses are changed; this is explicitly *not* a complete 89-constant re-pin.

The Higgs/top ratio moves from

$$125.25/172.57 = 0.725792432057$$

to

$$125.13/172.60 = 0.724971031286.$$

Independent receipt

The full-shelf counts become (76, 33, 9, 5, 0); familywise $p = 0.0170$. After removing math constants, the counts are (49, 18, 4, 2, 0) with familywise $p = 0.0965$. After the seven-representative deduplication, the counts are (23, 9, 2, 0, 0) with familywise $p = 0.3470$.

Boundary

The partial re-pin can move individual nearest neighbours and even increase fine-tier counts. The autopsy verdict remains unchanged: no physical identification survives dependence controls.

16.16 G16 – Mass-ratio uncertainty propagation

COMPUTED

Derivation

For $r = m_H/m_t$ with independent quoted uncertainties,

$$\left(\frac{\sigma_r}{r}\right)^2 = \left(\frac{\sigma_H}{m_H}\right)^2 + \left(\frac{\sigma_t}{m_t}\right)^2.$$

Using the re-pinned values,

$$\frac{\sigma_r}{r} = 1.794397e - 03.$$

The relative offset between r and $2\pi/(5\sqrt{3})$ is

$$\delta = 7.568777e - 04.$$

Independent receipt

The input uncertainty is 2.371 times the advertised match offset.

Boundary

A coincidence finer than the uncertainty of the quantity it claims to identify cannot be tested at that advertised precision.

16.17 G17 – Schwinger contrast

EXACT

COMPUTED

Derivation

With $\alpha^{-1} = 137.035999177$,

$$a_e^{(1)} = \frac{\alpha}{2\pi} = 1.161409732098e - 03.$$

Against the measured anomaly

$$a_e = 1.159652180590e - 03,$$

the first-order relative difference is

$$1.515585e - 03.$$

Independent receipt

The one-line calculation reproduces the source output.

Boundary

This is a forward formula with no transform menu. Higher-order QED terms, independent input α , and precision measurements are what turn resemblance into theory testing.

16.18 G18 – Pre-registered spectral-dimension test

HYPOTHESIS

Derivation

The source seals a future calculation on a named generation-5 contact graph. The primary observable is the return probability

$$P_0(t) \sim t^{-d_s/2}, \quad d_s = -2 \frac{d \ln P_0}{d \ln t},$$

with an independent small-eigenvalue-density estimate from the graph Laplacian.

Independent receipt

No result is present in the supplied files, so this tower records the protocol but does not invent a value.

Boundary

The hypotheses $d_s < 2$ and $d_s \simeq 2$ remain open until the specified graph, walker count, time range, connectivity check, fit window, and independent spectral estimator are executed.

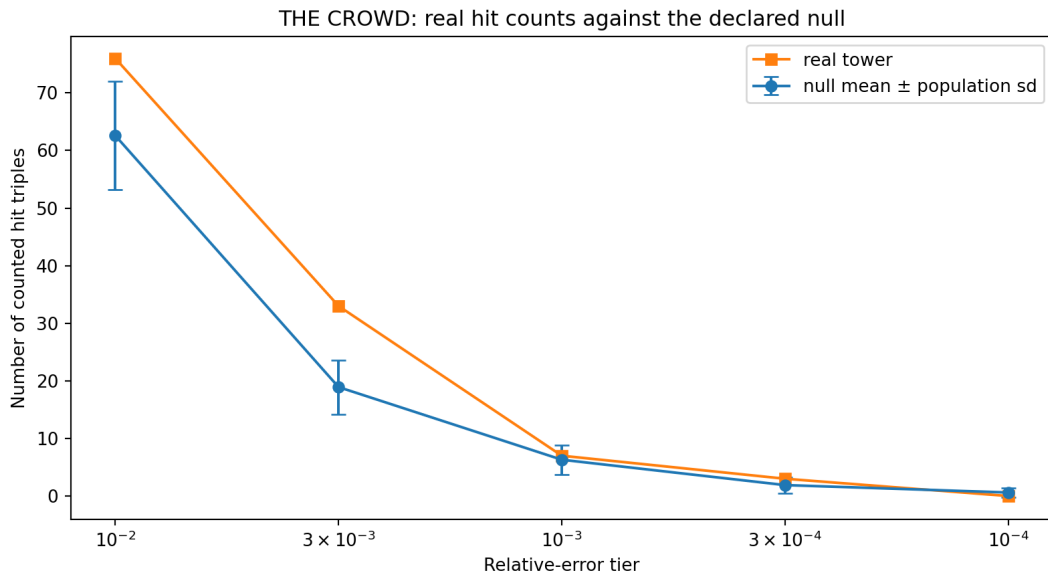


Figure 16.1: Original THE CROWD hit counts against the declared null.

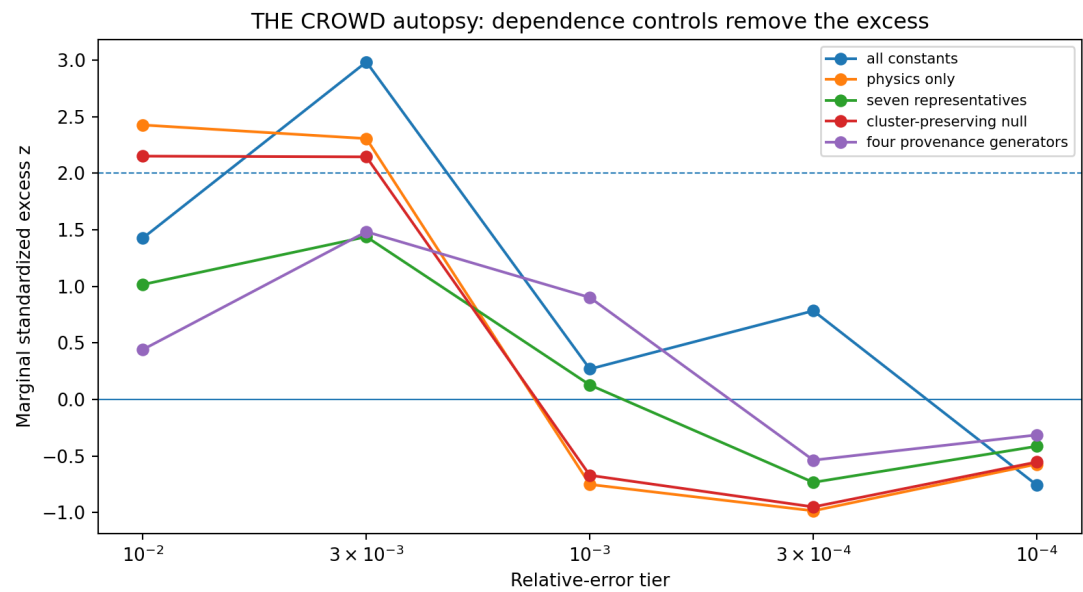


Figure 16.2: Marginal standardized profiles through the shared-alphabet, clustering, and provenance controls.

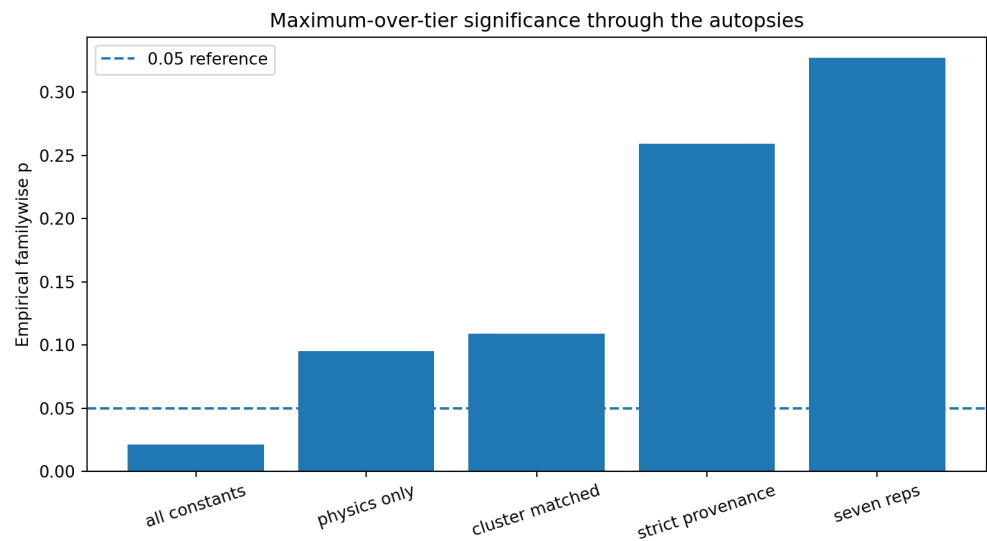


Figure 16.3: Empirical familywise p -values after taking the maximum over all five tiers.

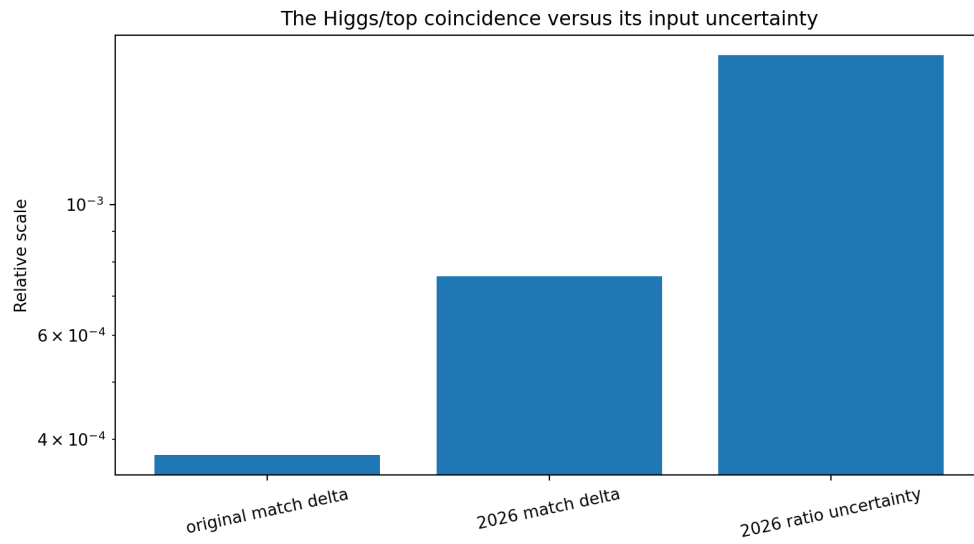


Figure 16.4: The Higgs/top nearest-neighbour offset remains smaller than the uncertainty of the ratio it is supposed to identify.

Chapter 17

v1.3.3 Calculation Receipt

Deterministic totals

Frozen core checks: 38/38.
Live Goldberg shell receipts: 8.
Live LUCA locks: 34/34 non-axiomatic rows.
Live LUCA EML price: 44,185 calls.
THE CROWD original tables: reproduced exactly.
Stable v1.3.3 payload SHA-256:
1e6c813e7f6ba8e00587836d2bbbed90
b5d7e8a1ee0ebba4343f17dbcf3e3cb7.

17.1 New executable files

```
python verify_light_matrix_v133_calculations.py
node verify_luca_v02_live.js
python build_light_matrix_v133_calculation_tower.py
latexmk -pdf THEA_LIGHT_MATRIX_v1.3.3_CALCULATION_TOWER.tex
```

Chapter 18

Independent Audit Receipt

The following table is generated from `light_matrix_v1.3.2_audit_receipt.json`. Long result fields are truncated here; the JSON preserves the full values.

#	check	status	verdict	result
1	Euler forces twelve pentagons	EXACT	PASS	{'forced_pentagons': 12, 'solution': {'E': '3*H + 30', 'P': '12', 'V': '2*H + 20'}}
2	C60 combinatorial counts from T=3	EXACT	PASS	{'E': 90, 'F': 32, 'H': 20, 'P': 12, 'T': 3, 'V': 60, 'chi': 2}
3	discrete Gauss-Bonnet charge	EXACT	PASS	4*pi
4	hexagonal closure determinant	EXACT	PASS	det M(k,l) = k^2 + k l + l^2
5	hexagonal metric similarity	EXACT	PASS	M^T Q M = T Q
6	closure matrix composition	EXACT	PASS	{'ell_prime': 'a*l + b*k + b*l', 'k_prime': 'a*k - b*l'}
7	hexagonal norm multiplicativity	EXACT	PASS	N(gz)=N(g)N(z)
8	leapfrog multiplier and count tower	EXACT	PASS	{'area_multiplier': 3, 'linear_multiplier': 1.7320508075688772, 'vertex_sequence': [20, 60, 180, 540, 1620]}
9	selected is not fixed golden nesting	EXACT / CORRECTION	PASS	{'argument': 'fixed GC area multiplier is an integer S; sqrt(S)=phi would require S=phi^2 irrational'}
10	Fibonacci selector pairs	EXACT	PASS	[[1, 0], [1, 1], [2, 1], [3, 2], [5, 3], [8, 5], [13, 8], [21, 13]]
11	projective golden error identity	EXACT	PASS	r_{n+1}-phi = -(r_n-phi)/(phi r_n)
12	golden triangulation sequence	EXACT	PASS	[1, 3, 7, 19, 49, 129, 337, 883, 2311, 6051]
13	golden T recurrence	EXACT	PASS	T[n+3] = 2T[n+2] + 2T[n+1] - T[n]
14	golden T closed form	EXACT	PASS	{'last': 91530451, 'verified_levels': 20}
15	T ratio alternates rather than climbs monotonically	EXACT / CORRECTION	PASS	{'first_ratios': [2.3333333333333335, 2.7142857142857144, 2.5789473684210527, 2.63265306122449, 2.612403100775194, 2.620178041543027], 'l...
16	golden-ray radical angle	EXACT	PASS	{'degrees': 22.23875609296496, 'difference_between_forms': 3.3306690738754696e-16, 'radians': 0.3881395153701887}
17	golden-ray angular contraction	EXACT + COMPUTED	PASS	{'last_reported_ratio': -0.3819660304017546, 'levels': [3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18], 'target': -0.3819660112...
18	symmetric-square light-matrix lift	EXACT	PASS	s(k+l,k,P)=M_light s(k,l,P)
19	light-matrix Faddeev-LeVerrier coefficient list	EXACT	PASS	[1, -3, 0, 3, -1]
20	light-matrix characteristic factorization	EXACT	PASS	(lambda - 1)*(lambda + 1)*(lambda**2 - 3*lambda + 1)
21	light-matrix anti-palindromy	EXACT	PASS	p(lambda) = -lambda^4 p(1/lambda)
22	light-matrix reciprocal spectrum	EXACT	PASS	[sqrt(5)/2 + 3/2, '1', '-1', '3/2 - sqrt(5)/2']
23	light-matrix eigenbasis	EXACT	PASS	{'-1': True, '1': True, 'phi^-2': True, 'phi^2': True}
24	C60 graph topology certificate	EXACT	PASS	{'E': 90, 'F': 32, 'V': 60, 'chi': 2, 'connected': True, 'degree_set': [3], 'face_counts': {'5': 12, '6': 20}}
25	C60 complete adjacency characteristic polynomial	EXACT	PASS	(lambda - 3)*(lambda - 1)**9*(lambda + 2)**4*(lambda**2 - lambda - 3)**5*(lambda**2 + lambda - 4)**4*(lambda**2 + lambda - 1)**5*(lambda*...
26	C60 minimum adjacency eigenvalue	EXACT + COMPUTED	PASS	{'exact': '-(3+sqrt(5))/2 = -phi^2', 'lambda_min': -2.618033988749895, 'multiplicity': 3}
27	C60 Fiedler radical and quartic	EXACT + COMPUTED	PASS	{'difference': 7.882583474838611e-15, 'numeric': 0.2434017461399247, 'quartic_residual': 8.926193117986259e-14, 'radical': 0.243401746139...
28	C60 low-band multiplicities	COMPUTED	PASS	[{'multiplicity': 3, 'value': 0.2434017461399247}, {'multiplicity': 5, 'value': 0.697224362268004}, {'multiplicity': 3, 'value': 1.179750...}
29	continuum coefficient 2pi/(5sqrt3)	EXACT CONDITIONAL / CORRECTION	PASS	{'coefficient': '2*sqrt(3)*pi/15', 'numeric': 0.7255197456936872, 'status': 'conditional asymptotic prediction, not a proved discrete gra...

#	check	status	verdict	result
30	SU3 Casimir and dimension formulas	EXACT	PASS	{'C2': 'p**2/3 + p*q/3 + p + q**2/3 + q', 'dimension': '(p/2 + 1/2)*(q + 1)*(p + q + 2)', 'table': {'(0, 1)': 3, '(1, 0)': 3, '(1, 1)': 8...
31	A2 root/weight lattice index-three relation	EXACT / CORRECTION	PASS	{'Cartan_matrix': [['2', '-1'], ['-1', '2']], 'correction': 'dual/commensurable hexagonal lattices, not literally identical', 'determinan...
32	Klein icosahedral syzygy	EXACT	PASS	T_30^2 + H_20^3 = 1728 f_12^5 (symbolically zero residual)
33	Klein gradient equivariants	EXACT	PASS	{'g11': {'degree': 11, 'projective_gradient': True, 'scale': '1'}, 'g19': {'degree': 19, 'projective_gradient': True, 'scale': '1/20'}, ...
34	float64 exactness break indices	COMPUTED / CORRECTION	PASS	{'E_last_exact': 35, 'T_last_exact': 38, 'V_last_exact': 36, 'chi_first_bad_level': 36, 'chi_first_bad_value': 0.0, 'chi_last_correct': 35}
35	forward stability and backward instability	COMPUTED / CORRECTION	PASS	{'backward_relative_errors': {'0': 1.1864584222679098e+32, '10': 4.9512758706203005e+23, '30': 9455846.679125344, '50': 1.807821865712679...
36	binary precision and T147 bit budget	EXACT + COMPUTED / CORRECTION	PASS	{'T147_bit_length': 205, 'correction': 'binary256 has 237 significand bits, not a 256-bit mantissa', 'format_precision_bits': {'binary128...
37	EML positive-real identity and branch ledger	EXACT CONDITIONAL / CORRECTION	PASS	{'Log(exp(z))-z': '-6.283185307179586j', 'boundary': 'node identities require domain and branch labels; 16-point locks are numerical siev...
38	LUCA leaf regression, Vogel sweep, and Fibonacci fingerprint	COMPUTED / REFUTED / CORRECTION	PASS	{'finite_N_verdict': 'golden angle is not the exact N=33 grid optimum', 'golden_nearest_index_gaps': {'13': 11, '3': 2, '5': 4, '8': 16},...

Chapter 19

Reproduction Protocol

19.1 Commands

```
python verify_light_matrix_v132.py
python verify_light_matrix_v133_calculations.py
node verify_luca_v02_live.js
node verify_light_matrix_v131_live.js > light_matrix_v1.3.1_live_kernel_receipt.json
pdflatex THEA_LIGHT_MATRIX_v1.3.3_CALCULATION_TOWER.tex
pdflatex THEA_LIGHT_MATRIX_v1.3.3_CALCULATION_TOWER.tex
```

19.2 Version discipline

The new document does not overwrite any input. The source HTML and LUCA versions remain immutable. v1.3.3 is a derived calculation audit artifact whose corrections are explicit, whose receipts are machine-readable, and whose build timestamp is not included in the mathematical hash.

19.3 Closing line

The pentagons hold. The hexes pay.

The ray now has a bearing.

The limit still owes its proof.

$$P = 12. \quad \chi = 2. \quad \theta_\varphi = \arctan(\sqrt{15} - 2\sqrt{3}).$$

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